计算机网络-题库

A ⟶ B

1、How long does it take to send a file of 640,000 bits from host A to host B over a circuit-switched network? All links are 1.536 Mbps, Each link uses TDM with 24 slots, 500 msec to establish end-to-end circuit.

时分复用 $v=\frac{1536}{24}=64000\,bps$ $t=\frac{64000}{64000}=1s$ $t_{总}=0.5+10=10.5$

答案:

解：1536000/24=64000bps 640000/64000 = 10s 10+0.5 = 10.5s

2、Assuming that the link length d between two hosts is 60m, the link bandwidth R is 10Mbps, and the signal propagation rate s is $2x10^8$ m / s. Now host A sends 1-bit packet to host B.

$\frac{60m}{2\times10^8}=3\times10^{-7}s$ $\frac{1}{10M}=10^{-7}s$

t_{trans} ↳传输时延

t_{prop} ↳传播时延

t=传播时延+传输时延 $=4\times10^{-7}s$.

1) How long does it take for host B to receive this packet? If host A starts transmitting the packet at time t = 0, how far does the front end of the signal of the packet spread at time t = d_{trans}?

当$t=1\times10^{-7}s$时 $S=2\times10^8\times1\times10^{-7}=20m$.

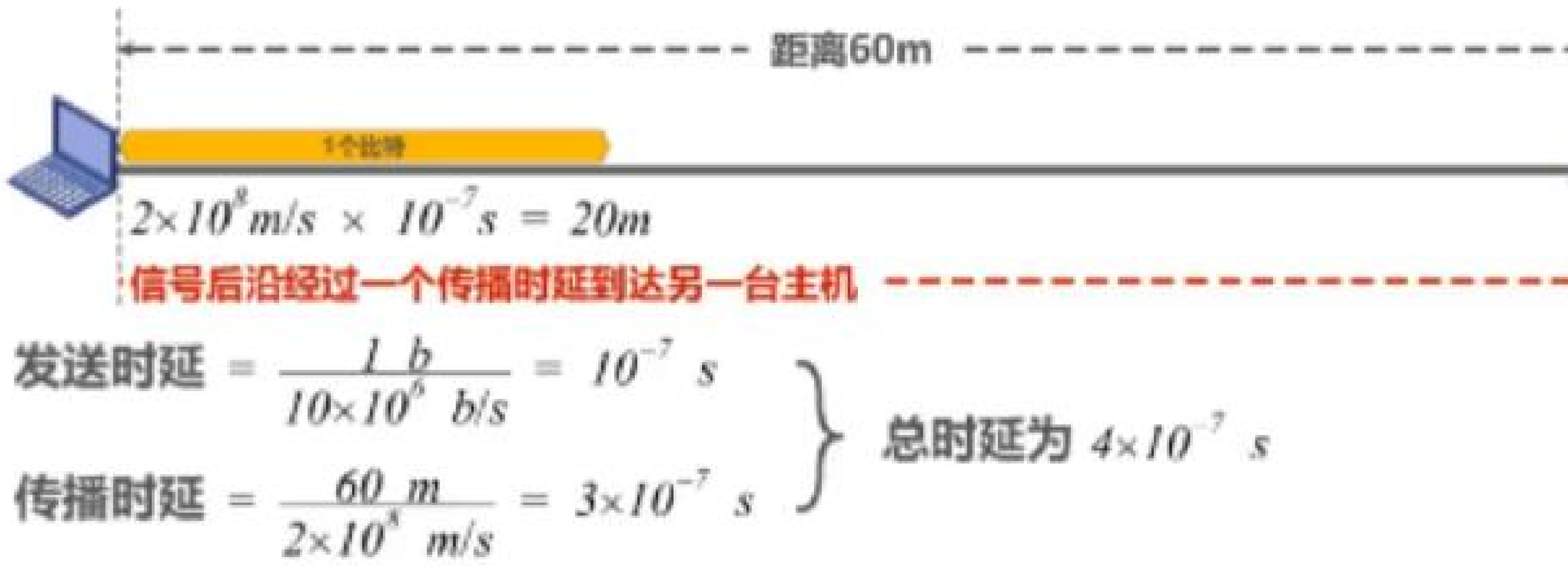
2) What if the link length d is 20 meters?

答案:

1)

$trans=\frac{1}{10^7}=1\times10^{-7}s$

$tprop=\frac{20}{2\times10^8}=1\times10^{-7}s$ $\Rightarrow t=2\times10^{-7}s$



t = d_{trans} 时，信号前端在 20m 处

2）



$$\text{发送时延} = \frac{1\ b}{10\times10^{6}\ b/s} = 10^{-7}\ s$$

$$\text{传播时延} = \frac{20\ m}{2\times10^{8}\ m/s} = 10^{-7}\ s$$

} 总时延为 2×10^{-7} s

带宽是用来计算传输时延 ttrans 的，相当于传输能力的速率

3、Assuming that the link length d between two hosts is 60m, the link bandwidth R is 10Mbps, and the signal propagation rate s is 2×10^{8} m / s. Now host A sends n-bit packet to host B, how long does it take for host B to receive the packet?

答案：

$(n+3)\times10^{-7}$ s

$t_{trans} = \frac{n}{10M} = n\times10^{-7}s$

$t_{prop} = \frac{s}{v} = \frac{60}{2\times10^{8}} = 3\times10^{-7}s$

$t = (n+3)\times10^{-7}s$

解析：

$$\text{发送时延} = \frac{1\ b}{10\times10^{6}\ b/s} = 10^{-7}\ s$$

$$\text{传播时延} = \frac{60\ m}{2\times10^{8}\ m/s} = 3\times10^{-7}\ s$$

} 总时延为 4×10^{-7} s

接收完n比特的总时延为 $(4\times10^{-7}s)\times n$ ✗

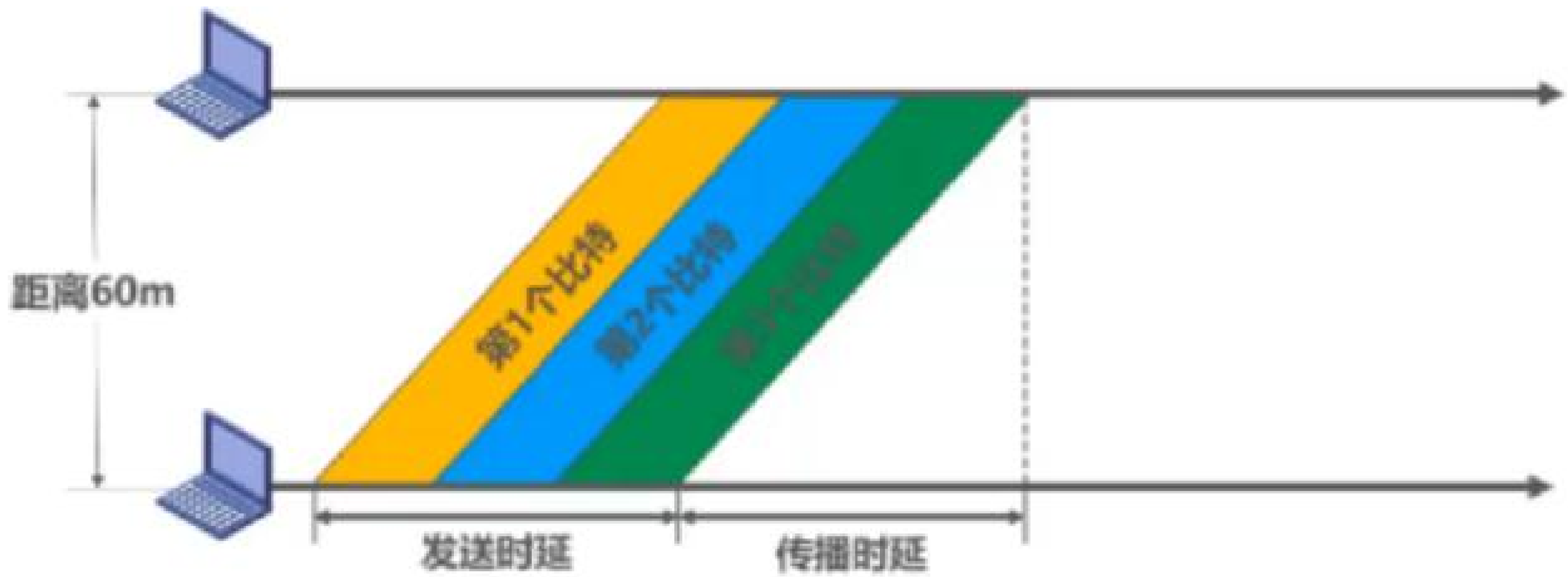
$$\text{发送时延} = \frac{n\ b}{10\times10^{6}\ b/s} = n\times10^{-7}\ s$$

$$\text{传播时延} = \frac{60\ m}{2\times10^{8}\ m/s} = 3\times10^{-7}\ s$$

} 总时延为

第一种计算方法是错误的。比特信号是一个跟着一个向前同时传播的，因此，无论有多少比特，在总时延中只包含一个传播时延。增加比特数量，只是增大了发送时延，而传播时延数量不变。

结论：若主机在一段链路上连续发送分组，则总时延为所有分组的发送时延加上信号在这一段链路上的传播时延。



主机A向主机B发送n个数据包，每个数据包的大小为L，两个主机之间有m条链路，每条链路长度为D，链路带宽为R → 传播时延 信号传播速率为S，忽略路由器的处理延时

4、Assuming that there m links between two hosts, each link length is D, the link bandwidth is R, and the signal propagation rate is S. Now host A sends n packet to host B, each packet size is L, how long does it take for host B to receive the packet? (Ignore the processing delay of the router)

$t_{trans} = \frac{nL}{R}$

$t_{prop} = \frac{mD}{S}$

$t = \frac{(m-1)D}{R}$

$\frac{nL}{R} + \frac{(m-1)D}{R} + \frac{mD}{S}$

答案:

n*L/R + (m−1)*L/R+m*D/S

解析:

把存储--转发的图画出来就清晰了。

若n个分组，m段链路，则总时延是多少?

5、中文教材 **P48**：**P10**，**P11**

P10. Consider a packet of length L that begins at end system A and travels over three lin
destination end system. These three links are connected by two packet switches. Let d_i, s
R_i denote the length, propagation speed, and the transmission rate of link i, for i=1,2,3. Th
packet switch delays each packet by d_{proc}. Assuming no queuing delays, in terms of d_i, s_i,
(i=1,2,3), and L, what is the total end-to-end delay for the packet? Suppose now the pack
1,500 bytes, the propagation speed on all three links is 2.5×10^8 m/s, the transmission rate
three links are 2 Mbps, the packet switch processing delay is 3 msec, the length of the fir
5,000 km, the length of the second link is 4,000 km, and the length of the last link is 1,000
For these values, what is the end-to-end delay?

要转化为bit为单位

P11. In the above problem, suppose R1=R2=R3=R and dproc=0. Further suppose the pa
switch does not store-and-forward packets but instead immediately transmits each bit it re
before waiting for the entire packet to arrive. What is the end-to-end delay?

答案：在链路中有在传输延迟和传播时延，在交换机处由于有储蓄转发有在处理时延.

P10、第一终端系统要求 L/R_1 将分组发送到第一链路；分组在 d_1/s_1 中的第一链路上传播；分组交换机增加 d_{proc} 的处理延迟；在接收
后，连接第一和第二链路的分组交换机要求 L/R_2 将该分组发送到第二链路；该分组通过 d_2/s_2 中的第二链路传播。同样，我们可以找到
和第三个链路：L/R_3、d_{proc}和d_3/s_3所造成的延迟。加上这五个延迟

链路上的时延：

交换机上的时延

$$D_{trans} = L/R_1 + L/R_2 + L/R_3$$

$$D_{prop} = d_1/s_1 + d_2/s_2 + d_3/s_3$$

$$D_{prop} = 2*d_{proc}$$

$$D = D_{trans} + D_{prop} + D_{proc}$$

$$D = 6 + 6 + 6 + 20 + 16 + 4 + 3 + 3 = 64ms$$

$t_{trans} = \frac{L}{R_1} + \frac{L}{R_2} + \frac{L}{R_3}$

$t_{prop} = \frac{d_1}{s_1} + \frac{d_2}{s_2} + \frac{d_3}{s_3}$

$d_{proc} = 3ms \times 2$

$t_{总} = 6 + \frac{1500\times8}{2\times10^6}\times3 + \frac{10^7}{2.5\times10^8}$

$= 6 + 40 + 18 = 64ms$

P11、由于比特是立即发送的，所以分组交换机不引入任何延迟；特别是，它不引入传输延迟。因此，

$$d_{end2end} = L/R + d_1/s_1 + d_2/s_2 + d_3/s_3$$

假如bit立即发送 则不有在传输延迟

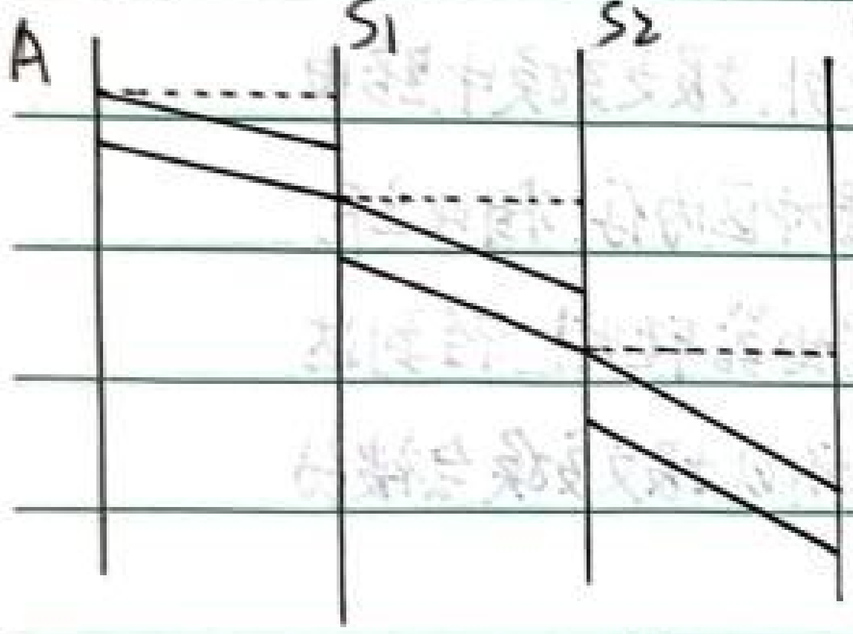
对于问题10中的值，我们得到

$$D = D_{prop} + L/R = 20 + 16 + 4 + 6 = 46ms$$

$t = 6 + 40 = 46$ ms.

解析：

5. P10



忽略排队时延的情况下：

端到端的延时为：

$$t_{delay} = t_{prop} + t_{trans} + 2t_{proc}$$

$$= \frac{d_1}{s_1} + \frac{d_2}{s_2} + \frac{d_3}{s_3} + \frac{L}{R_1} + \frac{L}{R_2} + \frac{L}{R_3} + 2t_{proc}$$

$L = 1500\ bytes = 1500 \times 8\ bits$　　代入到上式中可计算出

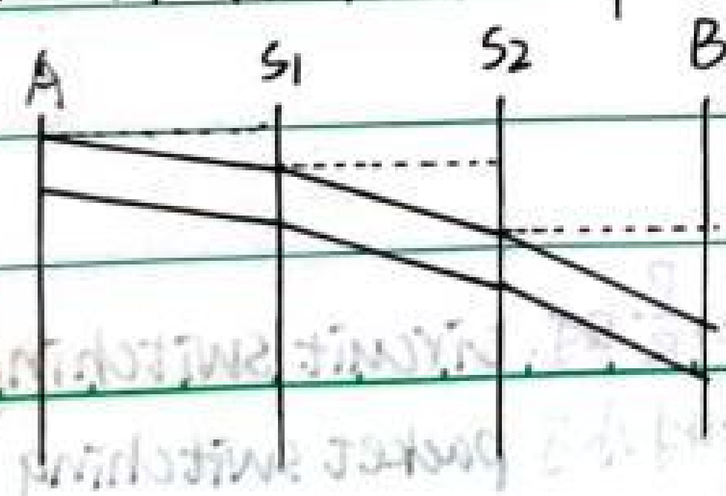
$R_1 = R_2 = R_3 = 2 \times 10^6\ bps$　　具体的端到端的延时为：

$t_{proc} = 3\ ms$　　$t_{delay} = 2 \times 3ms + \frac{12 \times 10^3\ bits}{2 \times 10^6\ bps} \times 3 + \frac{100 \times$ …

$d_1 = 5 \times 10^6 m$　$d_2 = 4 \times 10^6 m$　$d_3 = 1 \times 10^6 m$　$t_{delay} = 6ms + 18ms + 40ms$

$s_1 = s_2 = s_3 = 2.5 \times 10^8\ m/s$　　$t_{delay} = 64ms$.

P11. $R_1 = R_2 = R_3 = R$　$d_{proc} = 0$. 收到比特后立即转发，不进行存储.



由于 $R_1 = R_2 = R_3 = R$，故传输延时都相同

因此 $t_{delay} = \frac{L}{R} + \frac{d_1}{s_1} + \frac{d_2}{s_2} + \frac{d_3}{s_3}$

代入具体值可计算得：

$t_{delay} = 40ms + 6ms = 46ms$

6、Under the condition of no noise, if the frequency bandwidth of a communication link is 6kHz and QAM modulation technology with 4 phases and 2 amplitudes in each phase is adopted, What is the maximum data transmission rate of the communication link?

答案：在无噪声条件下，带宽为6kHz，采用QAM调制技术，

36Kbps 有4个相位和每个相位2个幅度

奈奎斯特公式　　$C = 2 \times 6000 \times \log_2(8) = 36\ kbps$.

$C = 2B\log_2(M)$

B-带宽

M-调制信号状态数

$M = 4 \times 2 = 8$

解析：波特率 6K，每个波特表示 3bit。C=2Wlog2N=2 * 6K * log2（4*2）=36Kbps

7、Suppose users share a 2 Mbps link. Also suppose each user transmits continuously at 1 Mbps when transmitting, but each user transmits only 20 percent of the time. (See the discussion of statistical multiplexing in Section 1.3.)

a. When circuit switching is used, how many users can be supported?

b. For the remainder of this problem, suppose packet switching is used. Why will there be essentially no queuing delay before the link if two or fewer users transmit at the same time? Why will there be a queuing delay if three users transmit at the same time?

c. Find the probability that a given user is transmitting.

d. Suppose now there are three users. Find the probability that at any given time, all three users are transmitting simultaneously. Find the fraction of time during which the queue grows.

答案：

a) 2 users can be supported because each user requires half of the link bandwidth.

b) Since each user requires 1Mbps when transmitting, if two or fewer users transmit simultaneously, a maximum of 2Mbps will be required. Since the available bandwidth of the shared link is 2Mbps, there will be no queuing delay before the link. Whereas, if three users transmit simultaneously, the bandwidth required will be 3Mbps which is more than the available bandwidth of the shared link. In this case, there will be queuing delay before the link.

c) Probability that a given user is transmitting = 0.2

d) Probability that all three users are transmitting simultaneously = $\binom{3}{3} p^3 (1-p)^{3-3}$ = $(0.2)^3 = 0.008$. Since the queue grows when all the users are transmitting, the fraction of time during which the queue grows (which is equal to the probability that all three users are transmitting simultaneously) is 0.008.

8、A user can directly connect to a server through either long-range wireless or a twisted-pair cable for transmitting a 1500-bytes file. The transmission rates of the wireless and wired media are 2 and 100 Mbps, respectively. Assume that the

propagation speed in air is 3×10^8 m/s, while the speed in the twisted pair is 2×10^8 m/s. If the user is located 1 km away from the server, what is the nodal delay when using each of the two technologies?

答案:

9、**P6. This elementary problem begins to explore propagation delay and transmis sion delay, two central concepts in data networking. Consider two hosts, A and B, connected by a single link of rate R bps. Suppose that the two hosts are separated by *m* meters, and suppose the propagation speed along the link is *s* meters/sec. Host A is to send a packet of size L bits to Host B.**

a. Express the propagation delay, d_{prop}, in terms of m and s.

b. Determine the transmission time of the packet, d_{trans}, in terms of L and R.

c. Ignoring processing and queuing delays, obtain an expression for the end to-end delay.

d. Suppose Host A begins to transmit the packet at time $t = 0$. At time $t = d_{trans}$, where is the last bit of the packet?

e. Suppose d_{prop} is greater than dtrans. At time $t = d_{trans}$, where is the first bit of the packet?

f. Suppose d_{prop} is less than dtrans. At time $t = d_{trans}$, where is the first bit of the packet?

g. Suppose s = 2.5 • 108, L = 120 bits, and R = 56 kbps. Find the distance m so that dprop equals dtrans.

答案:

a) $d_{prop} = m / s$ 秒.

b) $d_{trans} = L / R$ 秒.

c) $d_{end-to-end} = (m / s + L / R)$ 秒.

d) 该比特刚刚离开主机 A.

e) 第一个比特在链路中，尚未到达主机 B.

f) 第一个比特已到达主机 B.

g) $m = \frac{L}{R} s = \frac{120}{56 \times 10^3} \left(2.5 \times 10^8\right) = 536$ 千米.

共享 3×10^6 bps 带宽，每个用户需要 1.5×10^5 bps $n=\frac{3\times10^6}{1.5\times10^5}=20$

10、**P8. Suppose users share a 3 Mbps link. Also suppose each user requires 150 kbps when transmitting, but each user transmits only 10 percent of the time. (See the discussion of packet switching versus circuit switching in Section 1.3.)**

a. When circuit switching is used, how many users can be supported? $n=20$

电路交换　　分组交换

b. For the remainder of this problem, suppose packet switching is used. Find the probability that a given user is transmitting. 0.1

c. Suppose there are 120 users. Find the probability that at any given time, exactly n users are transmitting simultaneously. (Hint: Use the binomial distribution.)

d. Find the probability that there are 21 or more users transmitting simultaneously.

答案:

21及以上用户同时传输的概率

C. 二项分布 $P=C_{120}^{n} p^n (1-p)^{120-n}$ $p=0.1$

d. 20及以下个用户同时传输的概率

$P_1 = \sum_{n=1}^{20} C_{120}^{n} p^n (1-p)^{120-n}$

则 $P = 1-P_1 = 1-\sum_{n=1}^{20} C_{120}^{n} p^n (1-p)^{120-n}$ $p=0.1$

a) 能够支持 20 个用户.

b) $p = 0.1$.

c) $\binom{120}{n} p^n (1-p)^{120-n}$.

d) $1 - \sum_{n=0}^{20} \binom{120}{n} p^n (1-p)^{120-n}$.

利用中心极限定理来逼近这一概率，设独立随机变量为 $P(X_j =$

$P($ 21 个或更多用户同时传输 $) = 1 - P\left(\sum_{j=1}^{120} X_j \le 21\right)$

$$P\left(\sum_{j=1}^{120} X_j \le 21\right) = P\left(\frac{\sum_{j=1}^{120} X_j - 12}{\sqrt{120 \cdot 0.1 \cdot 0.9}} \le \frac{9}{\sqrt{120 \cdot 0.1 \cdot 0.9}}\right)$$

$$\approx P\left(Z \le \frac{9}{3.286}\right) = P(Z \le 2.74)$$

$$= 0.997$$

当为标准正态有效值时，$P($ 21 个或更多用户同时传输 $) \approx 0.00$

11、**P10. Consider the network illustrated in Figure 1.16. Assume the two hosts on the left of the figure start transmitting packets of 1500 bytes at the same time towards Router B. Suppose the link rates between the hosts and Router A is 4-Mbps. One link has a 6-ms propagation delay and the other has a 2-ms propagation delay. Will queuing delay occur at Router A?**

12000 bits.

4ms

$t = \frac{12000}{4 \times 10^6} = 3ms < 4ms$ 不会产生排队

答案:

端系统A需要 L/R_1 的时间将数据包传输到第一段链路上；数据包在第一段链路上传播的时间为 d_1/s_1；分组交换机增加了 d_{proc} 的处理时延；接收到整个包后，连接第一段链路和第二段链路的分组交换机需要 L/R_2 的时间将数据包传输到第二段链路上；数据包在第二段链路上传播的时间为 d_2/s_2。同样，还可以求出由第二个交换机和第三段链路引入的时延：L/R_3、d_{proc} 和 d_3/s_3。将所有时延相加得出

$$d_{end\text{-}end} = L/R_1 + L/R_2 + L/R_3 + d_1/s_1 + d_2/s_2 + d_3/s_3 + d_{proc} + d_{proc}$$

将题中数据带入式中可得到端到端时延为64 *msec*.

12、

流量强度 $I = \frac{La}{R}$

P14. Consider the queuing delay in a router buffer. Let I denote traffic intensity; that is, $I = La/R$. Suppose that the queuing delay takes the form IL/R for $I < 1$.

a. Provide a formula for the total delay, that is, the queuing delay plus the transmission delay.

b. Plot the total delay as a function of L/R.

答案:

a. 定义流量强度 $I = \frac{La}{R}$

则排队时延定义为 $t = \frac{IL}{R(1-I)}$ $(I<1)$

a) 传输时延为 L/R.

总时延为 $\frac{IL}{R(1-I)} + \frac{L}{R} = \frac{L/R}{1-I}$

$t_{trans} = \frac{L}{R}$

$t = t_{trans} + t_{排队} = \frac{IL}{R(1-I)} + \frac{L}{R} = \frac{IL+(1-I)L}{R(1-I)} = \frac{L}{R(1-I)}$

b. 绘制总延迟作为 $\frac{L}{R}$ 的函数

b) 设 $x = L/R$.

$t = \frac{L}{R(1-I)} = \frac{1}{1-I} \cdot (\frac{L}{R})$ 随着 $\frac{L}{R}$ ↑ t ↑

总时延 $= \frac{x}{1-ax}$

x = 0 时，总延迟为 0；随着 x 的增大，总时延增大，当 x 接近 1 时，总时延接近无穷大。

13、A LAN adopts CSMA/CD protocol to realize the media access control. The data transmission rate is 10Mbps, the distance between host A and host B is 2km, and the signal propagation speed is 200000km/s. If there is a conflict when two hosts send data,

A 采用 CSMA/CD

$t_{prop} = \frac{2\times10^3}{2\times10^8} = 10^{-5}s = 10\mu s$

$t_{trans} = \frac{data}{10^7}$

how long will it take from the time when they start sending data to the time when both hosts detect the conflict? (需要回答的问题是： What is the minimum and maximum duration?)

答案：

① 最短：A、B同时发送 $t_{min} = 10\mu s$

② 最长：A即将到达B时开始发 $t_{max} = 2 \times 10\mu s = 20\mu s$

(1) 主机甲和主机乙之间单向传播延迟时间：10μs。两台主机均检测到冲突时，
需时间和最长所需时间对应下面两种极端情况：

①主机甲和主机乙同时各发送一个数据帧，信号在信道中发生冲突后，冲突信号
两个方向传播。因此，甲乙两台主机均检测到冲突时，最短需经过10μs。

②主机甲(或主机乙)先发送一个数据帧，当该数据帧即将到达主机乙(或主机甲)时
机乙(或主机甲)也开始发送一个数据帧。这时，主机乙(或主机甲)将立即检测到冲突;
甲(或主机乙)要检测到冲突，冲突信号还需要从主机乙(或主机甲)传播到主机甲(或主
因此，甲乙两台主机均检测到冲突时，最长需经过20μs。

解析：

第五次作业.

1. the distance between A and B is 2 km $d = 2\,km$

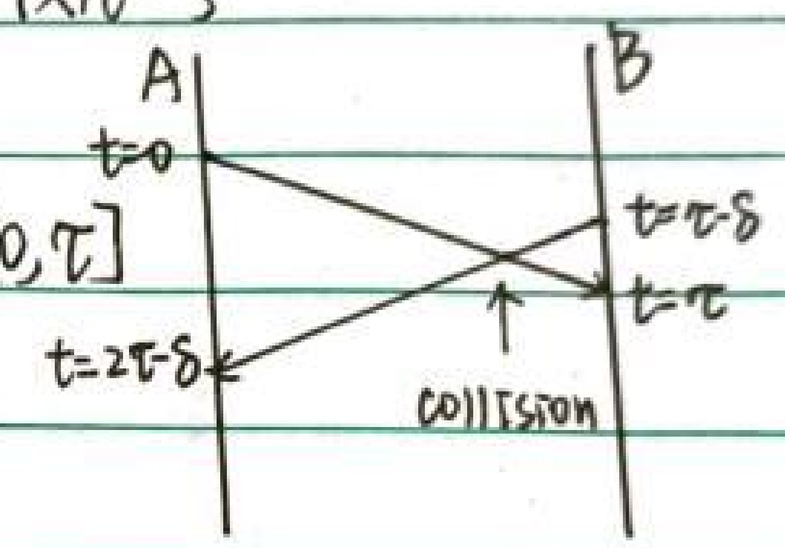
the signal propagation speed is 200000 km/s $S = 2\times10^5\ km/s$

$\therefore t_{prop} = \frac{d}{s} = 1\times10^{-5}\ s$

即 One way propagation delay : $\tau = 1\times10^{-5}\ s$

for B: $t=\tau$ 时可检测出冲突.

for A: $t = 2\tau - \delta$ 时可检测出冲突. $\delta \in [0, \tau]$



当 $\delta \to 0$. the maximum duration

is $2\tau = 2\times10^{-5}\ s$

当 $\delta \to \tau$. the minimum duration is $\tau = 1\times10^{-5}\ s$

14、

R6. In CSMA/CD, after the fifth collision, what is the probability that $K = 4$? The result $K = 4$ corresponds to a delay of how seconds on a 10 Mbps Ethernet?

$\frac{1}{2^5} = \frac{1}{32}$

$t_{等待} = 4\times512 = 2048\ \mu s$.

传输一位需要 $\frac{1}{10\times10^6} = 0.1\ \mu s$ 则 512 对应 $512\times0.1 = 51.2\ \mu s$

答案:

在第五次碰撞之后，适配器从{0, 1, 2, ..., 31}中选择 K=4 的概率为 1/32，需要等待 204.8 微秒。

15、

P5. Consider the generator, G = 1001, and suppose that D h value 11000111010. What is the value of R?

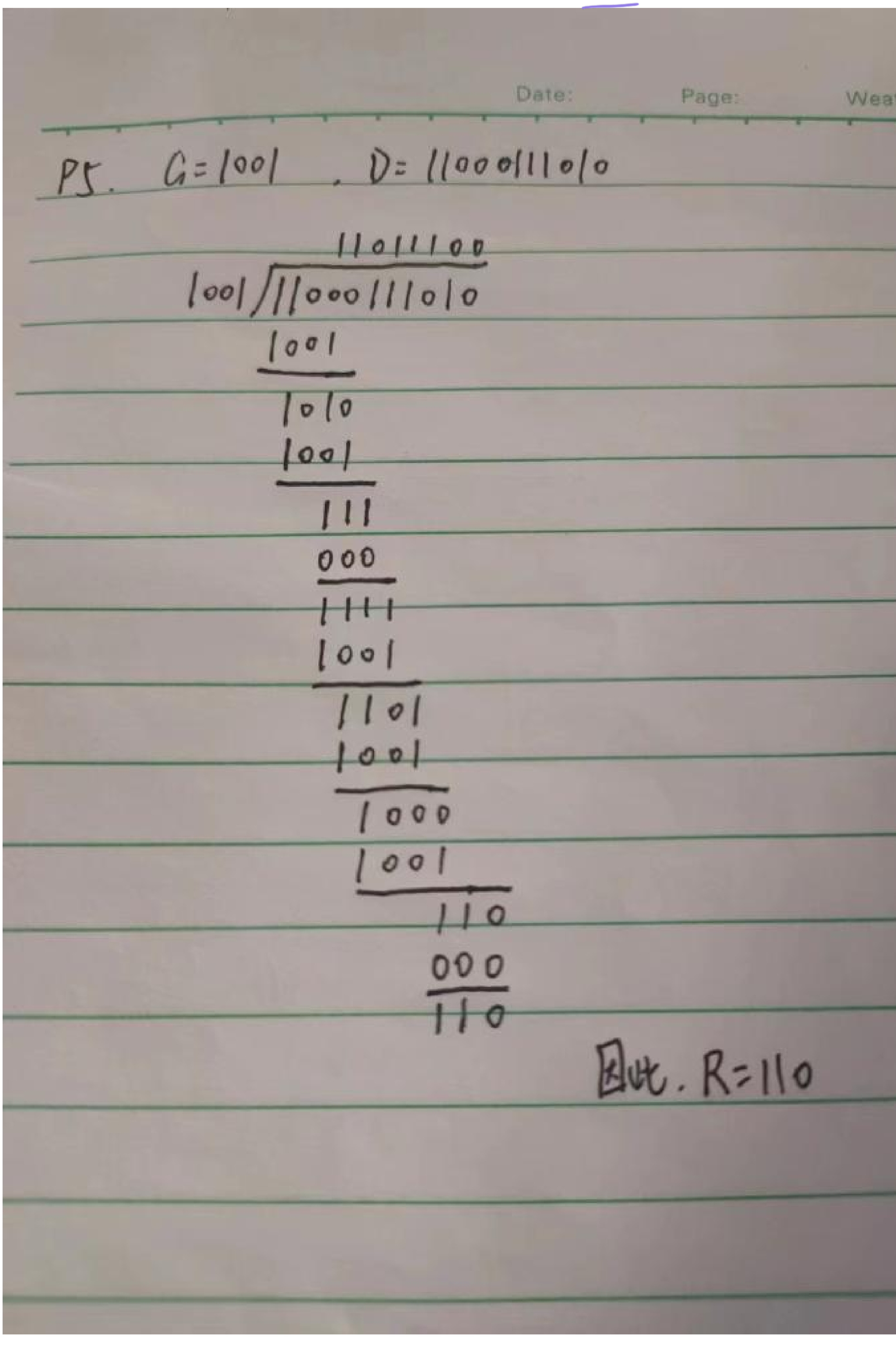
答案:

G=1001 (除数)

D=11000111010 (被除数)

```
        11011100
    ------------
1001| 11000111010
    1001
    ----
     1010
     1001
     ----
      1111
      1001
      ----
      1101
```

```
      1001
      1000
      1001
       110   R=110
```



Date: Page: Wea

PS. G=1001 , D=11000111010

```
          11011100
      ┌──────────
1001 ) 11000111010
       1001
       ────
        1010
        1001
        ────
         1110
         1001
         ────
          1111
          1001
          ────
           1100
           1001
           ────
            1010
            1001
            ────
             110
```

因此. R=110

16、

P6. Rework the previous problem, but suppose that D has the valu

a. 01101010101. R=011

b. 11111010101. R=011

c. 10001100001. R=110

答案：

a. R=011

b. R=011

c. R=110

17、

P8. In Section 6.3, we provided an outline of the derivation of the efficien slotted ALOHA. In this problem we'll complete the derivation.

时隙 ALOHA，效率公式 E= Np(1-p)^(N-1)，N表示活跃节点数

a. Recall that when there are N active nodes, the efficiency of slotted ALOHA is $Np(1-p)^{N-1}$. Find the value of p that maximizes this sion. 求使效率最大化的p值 当 $p=\frac{1}{N}$ 时效率最大化

b. Using the value of p found in (a), find the efficiency of slotted AL by letting N approach infinity. *Hint*: $(1-1/N)^N$ approaches $1/e$ as approaches infinity. E= 36.8%.

答案：

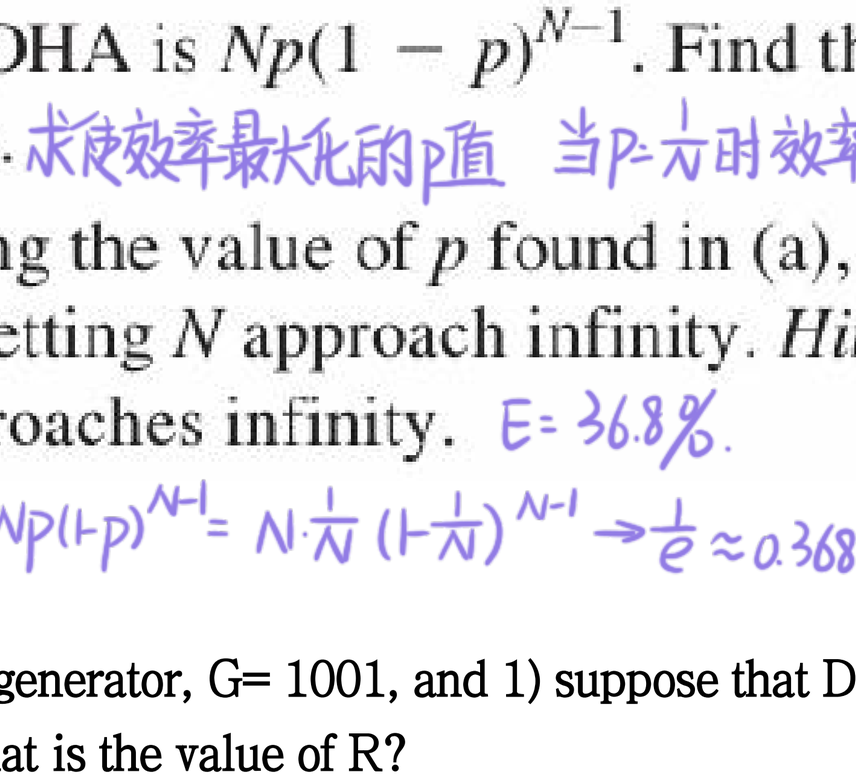
$E=Np(1-p)^{N-1}=N\cdot\frac{1}{N}\left(1-\frac{1}{N}\right)^{N-1}\to\frac{1}{e}\approx 0.368$.

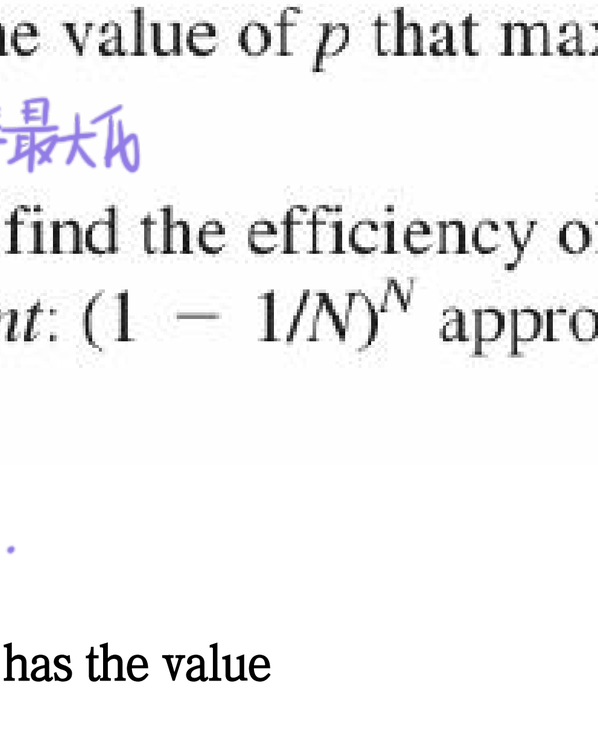
18、Consider the generator, G= 1001, and 1) suppose that D has the value 11000111010. What is the value of R?

2) if D has the value 11111010101. What is the value of R?

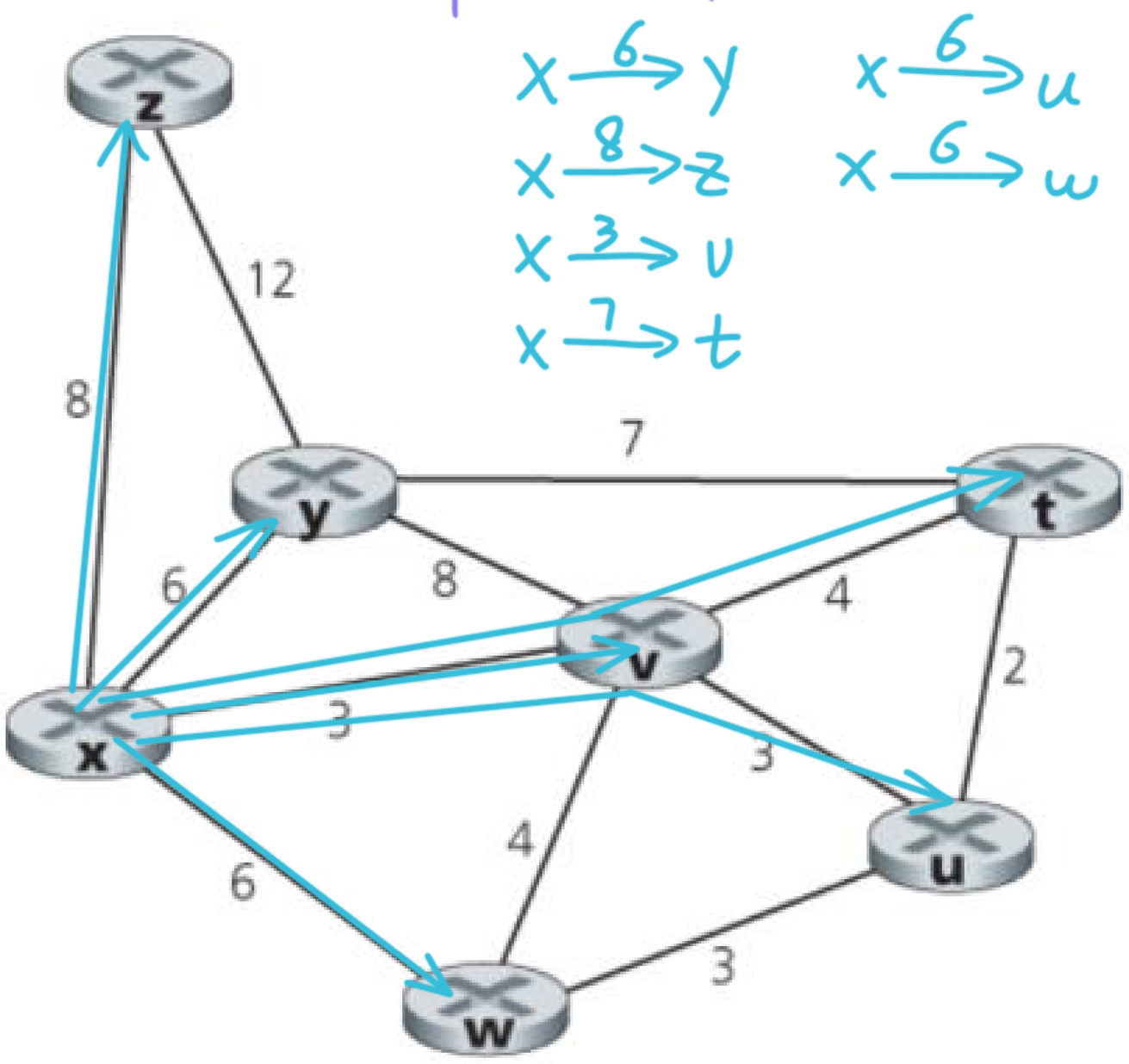
答案：

1) 110 2) 011



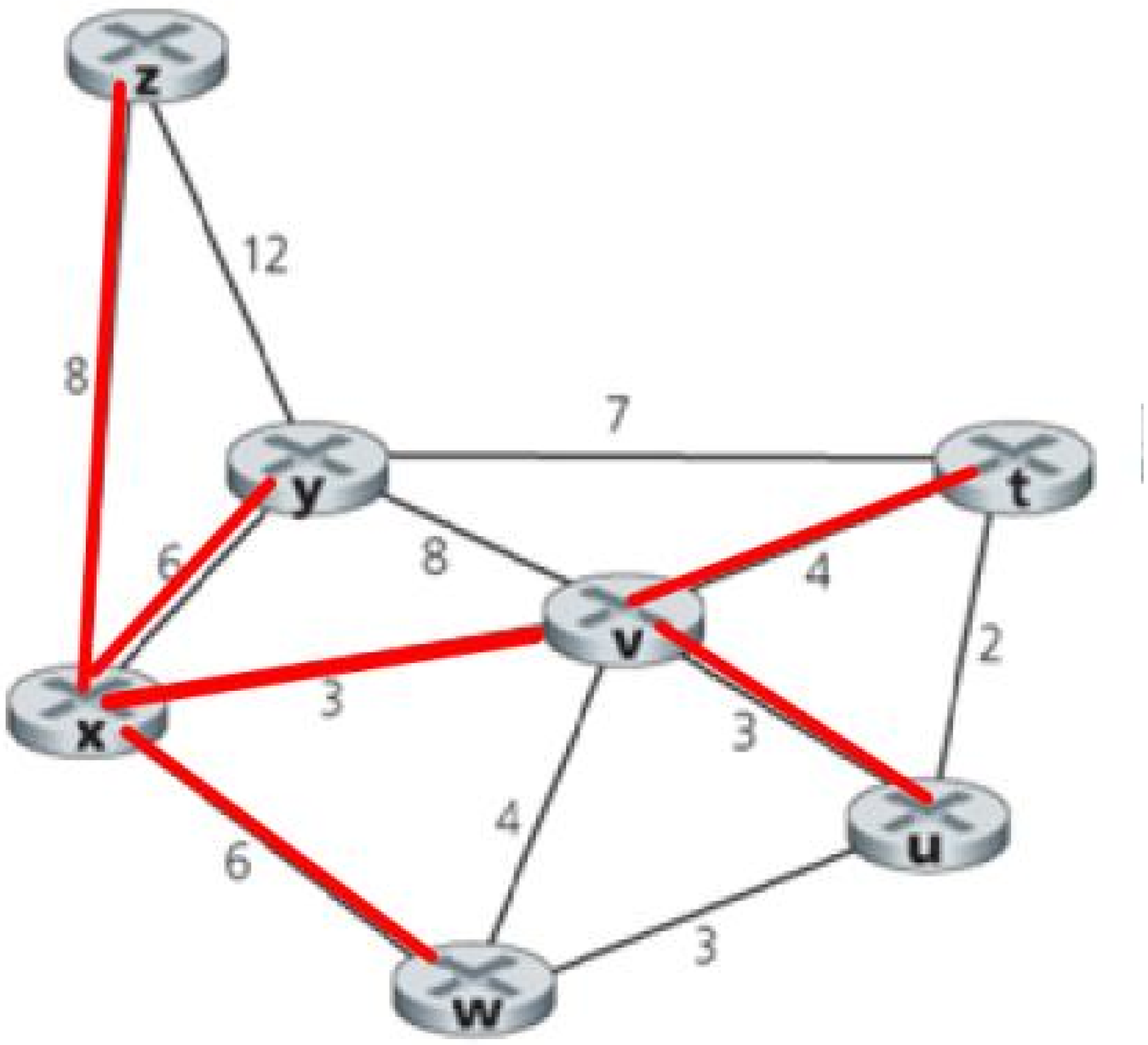


19、**Consider the following network. With the indicated link costs, please use Dijkstra's shortest-path algorithm to compute the shortest path from x to all network nodes, give out the shortest-path tree and the forwarding table from x to all network nodes.**

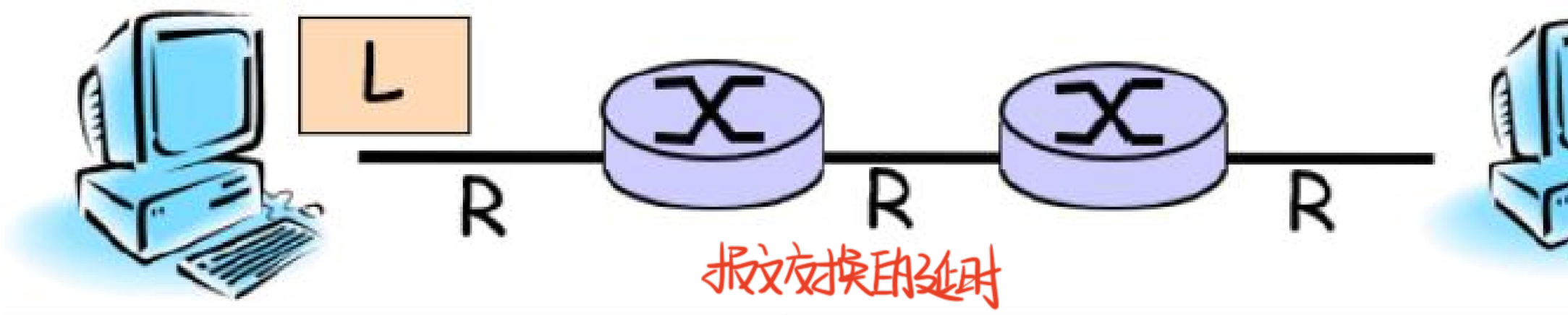


答案:

计算过程参考 PPT 或者教材;



20、



L = 7.5 Mbits

R = 1.5 Mbps $t_{trans} = \frac{7.5\times10^6}{1.5\times10^6} = 5\,s$

t_{prop} : propagation delay of each link

报文交换的延时

t=(传输时延+传播时延)×链路数

分组交换的延时

t=第1个分组的总延时 (t_{trans}+t_{prop}) + 剩余每个分组的传输时延

Divide the message into 5000 small packets: l=1.5 kbits

(1) Calculate the end-to-end delay of message switching（报文交换，数据不分组，直接发送） and packet switching（分组交换，数据分组后发送），

(2) Explain why the message switching is replaced by packet switching in the Internet.

(1) 报文交换 $t=(5+t_{prop})\times3=15+3t_{prop}$

分组交换 $t=(t_{prop}+\frac{1.5\times10^3}{1.5\times10^6})\times3+(5000-1)\times\frac{1.5\times10^3}{1.5\times10^6}=5.002+3t_{prop}$

答案：

message switching 报文交换，数据不分组，按单个 7.5M 数据发送

总延时：每次发送时间 X3+每段传播延时 x3　L/R*3+ 3*tprop

7.5/1.5+2*7.5/1.5+3*tprop=15+3*tprop

packet switching 分组交换，数据分组，按每个分组大小进行数据发送

总延时：分组数量 n=7.5M/1.5K = 5k　l/R*(n+2)+ 3*tprop

5000*1.5K/1.5M+2*1.5K/1.5M+3*tprop

=5.002+3*tprop



分组交换可以大幅减少发送时延



21、The message to be sent is X (bits) in total. There are K links from the source to the destination. The propagation delay of each link is D(s) and the bandwidth is R(bps). In circuit switching, the circuit establishment time is S(s). In packet switching, the message can be divided into several packets with P(bits) length. It is assumed that the processing time and queuing time of the message/packets at each node are ignored.

Q: under what conditions is the end-to-end delay of circuit switching greater than that of packet switching?

答案：

【答案】在电路交换的情况下，电路建立延迟等于 s 秒，发送延迟等于 x/b 秒，传播延迟等于 kd 秒。总延迟等于 $s + x/b + kd$ 秒。在分组交换的情况下，没有电路建立延迟，发送延迟（一个分组连续发送）也等于 x/b 秒，传播延迟等于 kd 秒，但增加了转发延迟 kp/b。总延迟等于 $x/b + kd + kp/b$ 秒。

令 $x/b + kd + kp/b < s + x/b + kd$，得到 $kp/b < s$。所以在 s 大于 kp/b 的条件下，分组交换网时延比电路交换网小。

解析：

3. 忽略处理时间和排队延时的情况下：

circuit switching：

$t_{delay} = t_{estab} + K t_{prop} + t_{trans}$

$= S + k \cdot D + \frac{x}{R}$

packet switching：

$t_{delay} = k \cdot t_{prop} + (k + \frac{x}{p} - 1) \cdot t_{trans}$

$= kD + (k + \frac{x}{p} - 1) \cdot \frac{P}{R}$

由题意：$S + kD + \frac{x}{R} < kD + k \cdot \frac{P}{R} + \frac{x}{R} - \frac{P}{R}$

$S < (k-1)\frac{P}{R}$，即 $S < (k-1)\frac{P}{R}$ 时，circuit switching 端到端的延时小于 packet switching

为确保冲突检测，CSMA/CD对最小帧长做3限制：

（最小帧的传输时间不小于往返时延）

☆ $L_{min} \geq R \cdot \frac{2D}{V}$ ⇒

L_{min}：最小帧长（bit）　V：信号在介质中的传播速度（m/s）

R：网络传输速率（bps）

D：两个最远节点之间的距离

22、

Ethernet is by far the most prevalent wired LAN technology, and it is likely to r
so for the foreseeable future. Ethernet uses CSMA/CD（载波监听/冲突检测）m
access protocol with binary exponential backoff to address the collision proble
shared broadcast link.

a) In a network using CSMA/CD protocol, the transmission medium is a co
cable, the transmission rate is 1Gbps, and the speed of signal propagation
cable is 200000km/s. If the minimum length of the data frame is 1000 bits
should be the distance between the farthest two nodes? (*m*)

要求 $L > R \cdot \frac{2D}{V}$ 则 $D < \frac{LV}{2R} = \frac{10^3 \times 2 \times 10^8}{2 \times 1 \times 10^9} = 100\,m$

b) According to the binary exponential backoff algorithm(二进制指数退避
after the 11th collision, what is the probability that a node chooses K=5? The
K=5 corresponds to a delay of how many seconds on a 10Mbps Ethernet?

$2^{10} = 1024$　$P = \frac{1}{1024}$　$t = \frac{1}{10 \times 10^6} = 10^{-7} = 0.1\mu s$

$t_{退} = 0.1 \times 512 \times 5 = 0.256\,ms$.

c) For 802.11 wireless LANs, the CSMA with collision avoidance mech
(CSMA/CA, 载波监听/冲突避免) is used instead of using collision detec
Ethernet. Please briefly describe how 802.11 implements the collision avoid

载波监听 → 空闲等待一个DIFS → RTS/CTS确保冲突避免 → ACK确认

答案：

(a) 100m

(b) 1/1024　　51.2us×5 = 256us

(c)

见第七版中文教材　P350-351

以及 https://blog.csdn.net/qq894040717/article/details/82426965

23、中文教材 **P47：R23**

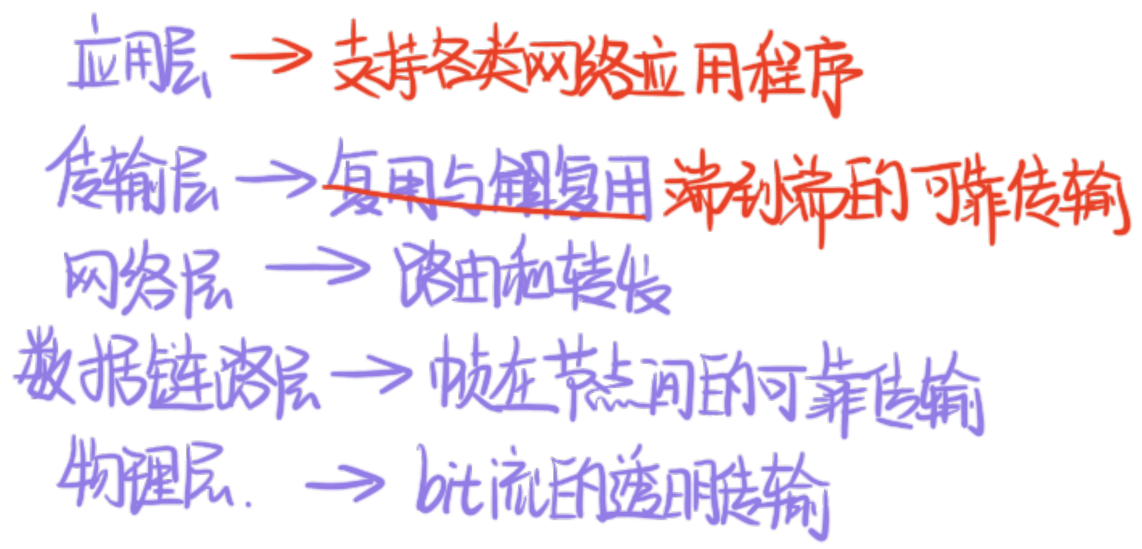
R23. What are the five layers in the Internet protocol stack? What are the prin
responsibilities of each of these layers?

答案：

网络协议的自顶向下的五层是：应用层、传输层、网络层、链路层和物理层。详见

1.

应用层：支持各类网络应用程序；



2.

3.

传输层：提供端到端（进程到进程）的可靠数据传输；

4.

5.

网络层：数据报路由和转发；

6.

7.

数据链路层：保证数据帧在相邻节点的可靠传输；

8.

9.

物理层：完成相邻节点链路上比特流的透明传输。

10.

4. ① application: supporting network applications

② transport: process-process data transfer

③ network: routing of datagrams from source to destin

④ link: data transfer between neighboring network elements

⑤ physical: bits on the wire

24、Please draw the signal waveform of 00110110 with NRZ-L, Manchester Coding, and Differential Manchester Coding.

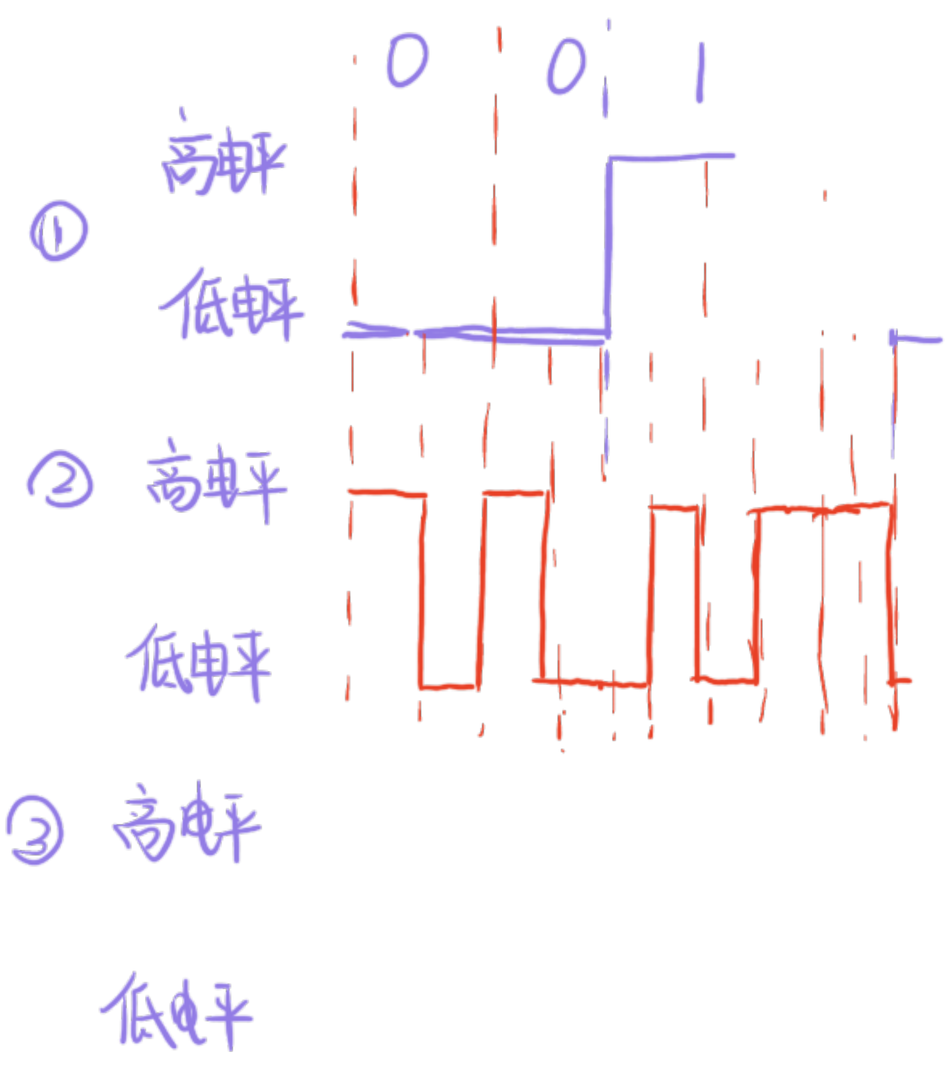
不归0编码

曼彻斯特编码 { 0是前半周期高电平后半周期低电平
1是前半周期低电平后半周期高电平

答案:

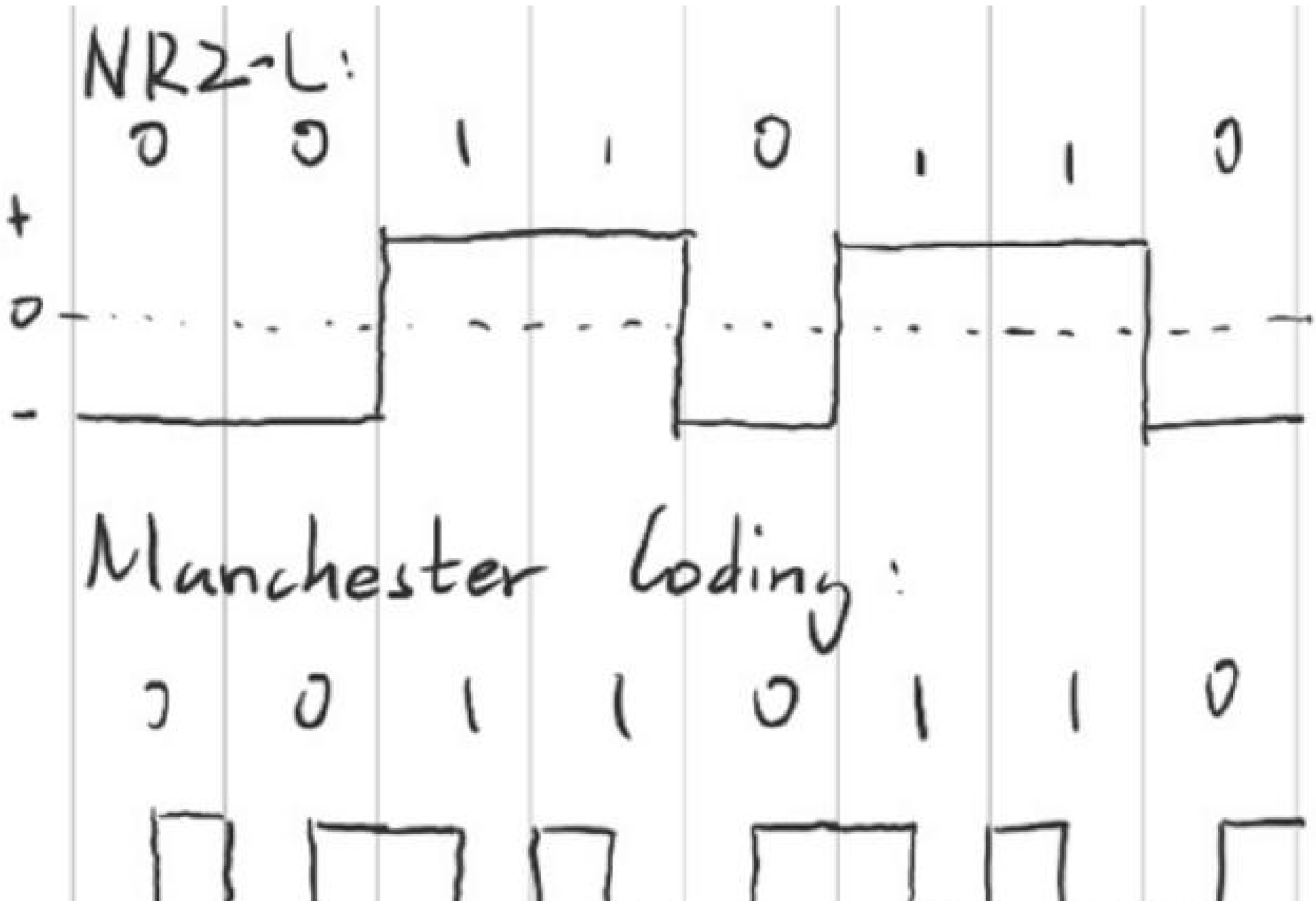
NRZ-L:

差分曼彻斯特编码：假设前一个码元的后半个波形是 高电平



差分曼彻斯特编码：每个码元周期的中间必须有一个跳变

{ 0：码元周期开始时发生跳变
1：码元周期开始时不发生跳变



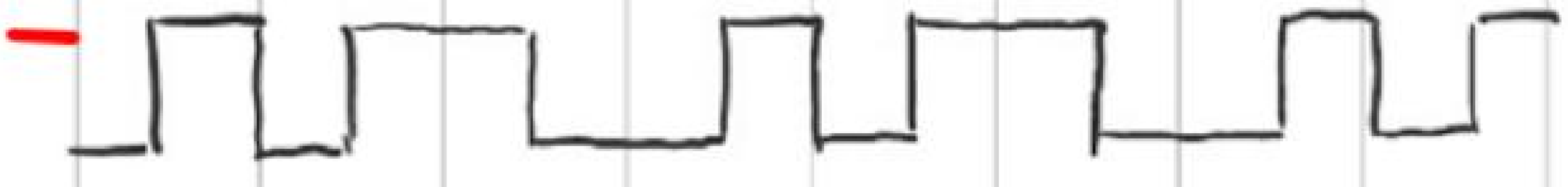
Manchester Coding:

0 0 1 1 0 1 1 0



Differential Manchester Coding:

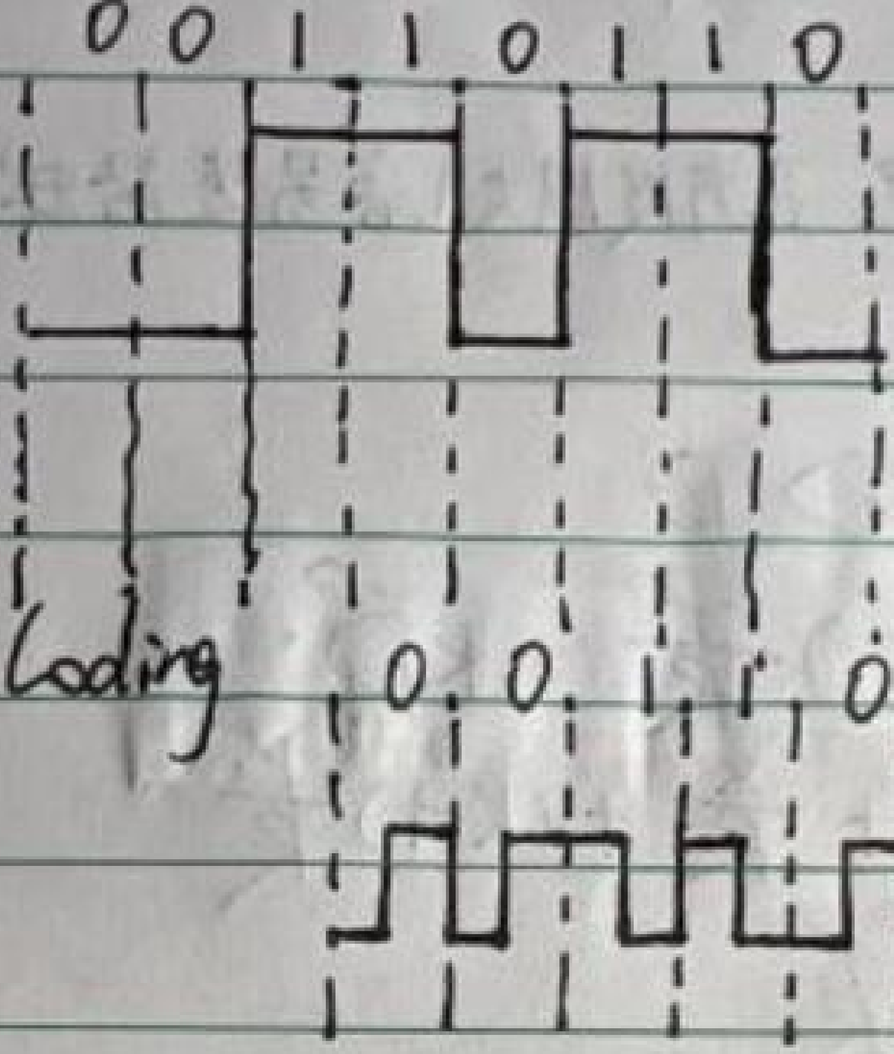
0 0 1 1 0 1 1 0



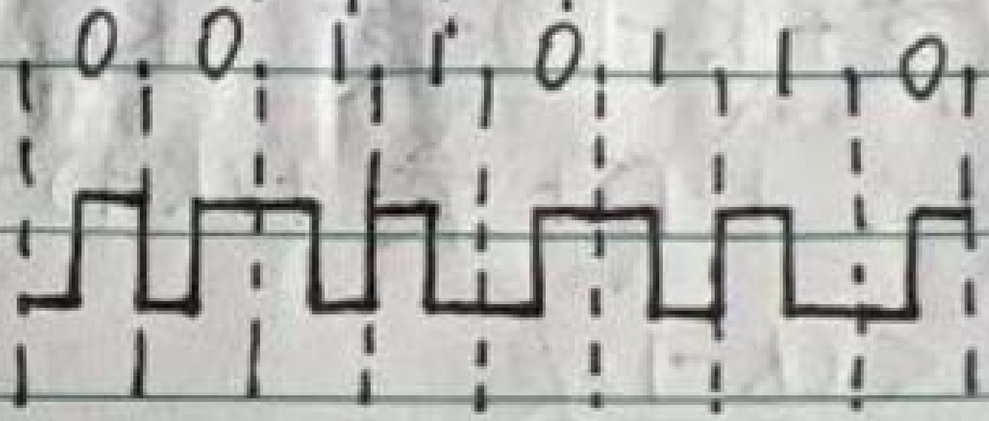
解析:

8. $C = 2W\log_2 N = 2 * 6k * \log_2(4*2) = 36Kbps$

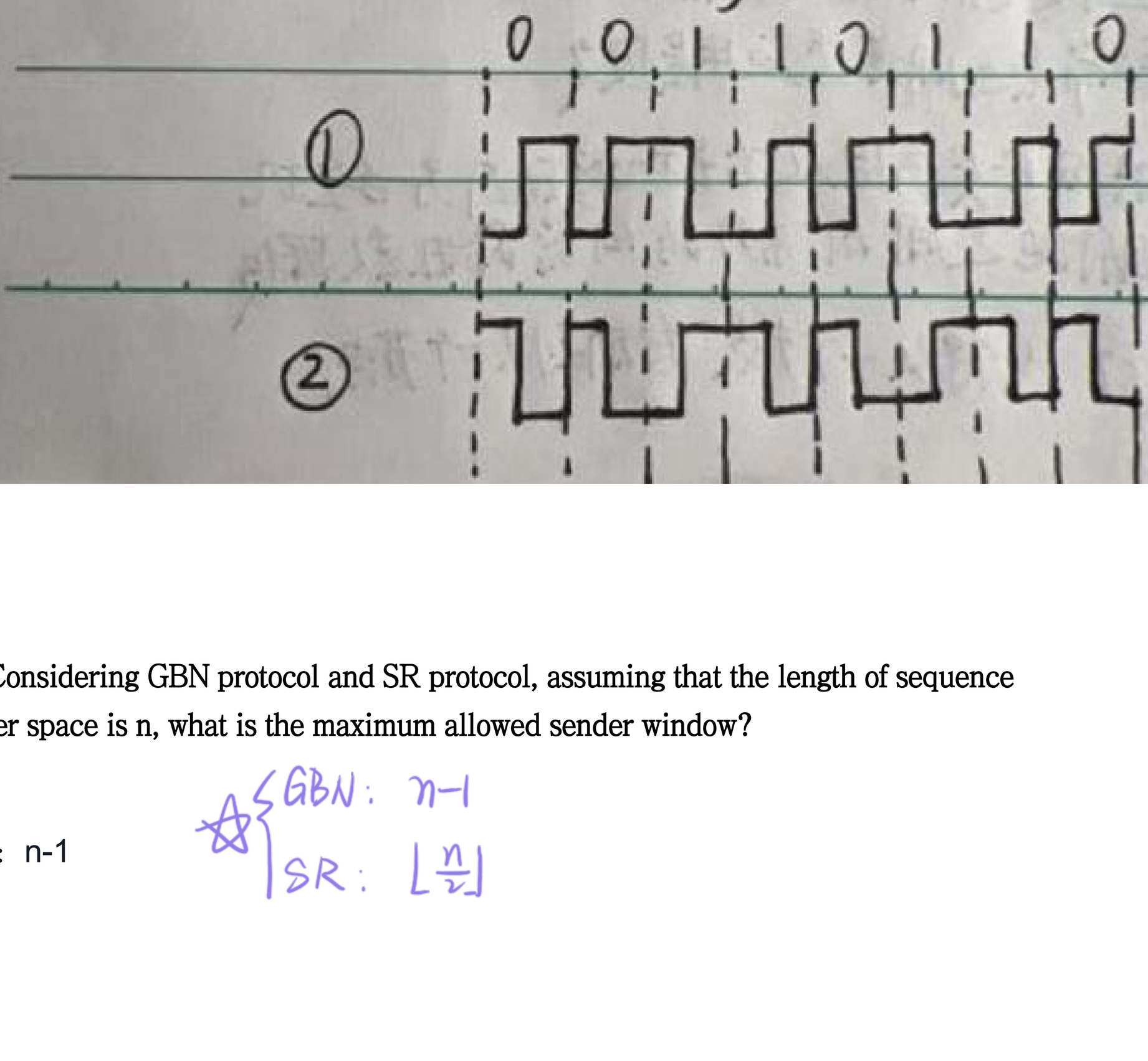
9. NRZ-L



Manchester Coding



Differential Manchester Coding



25、Considering GBN protocol and SR protocol, assuming that the length of sequence number space is n, what is the maximum allowed sender window?

答案:

GBN：n-1

☆ GBN: $n-1$

SR: $\lfloor \frac{n}{2} \rfloor$

SR：n/2

解析：

https://zhuanlan.zhihu.com/p/450499911

解析：如果发送方发送一整组数据，接收方接受数据并发出 ACK 消息，发送方此时没有接受到 ACK 消息；对于发送方而言，窗口为[m-w,m-1]；对于接收方而言，窗口为[m, m+w-1]；其中 m 表示 send_base，w 表示 window 大小；

问题：为了区分是第一个分组的重传还是下一组的初次传送

QUESTION：０１２３０１２３；序号 k 为 4；若此时窗口大小为 3

０１２３０１２３；[0, 1, 2]表示发送方窗口；[3, 0, 1]表示接收方窗口；若此时接收方接受到 0 分组，它为第一个分组的 0 的重传？还是下一个分组的 0 的初次传送?

QUESTION：０１２３０１２３；序号 k 为 4；若此时窗口大小为 2

０１２３０１２３；[0, 1]表示发送方窗口；[2, 3]表示接收方窗口；此时无论如何都不会出现相同序号的分组的情况。

即分组序号 k ⩾ 2w（窗口）.

26、

发送窗口 4 序列号范围1024

P22. Consider the GBN protocol with a sender window size of 4 and a sequence numbe
of 1,024. Suppose that at time *t*, the next in-order packet that the receiver is expecting h
sequence number of *k*. Assume that the medium does not reorder messages. Answer th
following questions:

(发送窗口内的可能序列号集合)

a. What are the possible sets of sequence numbers inside the sender's window at
Justify your answer. [k-x, k-x+3]

当前传播回发送方的ACK字段的可能值

b. What are all possible values of the ACK field in all possible messages currently
propagating back to the sender at time *t*? Justify your answer. k-3、k-2、k-1、k.

答案：

a ①所有packet都被确定 则(k, k+1, k+2, k+3)

②1个未确定， k-1, k, k+1, k+2

k-2, k-1, k, k+1

k-3, k-2, k-1, k.

a)这里我们的窗口大小为n=3。假设接收机接收到分组k-1,并且已经将该分组和所有其它之前的分组进行了分组。如果所有这些ACK都已被
收，则发送方的窗口为[k,k+n-1]。假设下一个没有在发送方接收到ACK。在第二种情况下,发送方S窗口包含k-1和n个分组，并包括k-1。因
口是这样的,[K-N,K-1]。通过这些参数，“发件人”窗口的大小为3，并开始在范围[K-N,K]的某处。

b)如果接收机正在等待分组k，则它已经接收（和被处理）分组k-1和N-1个分组。如果尚未收到这些N个ACK中的任何一个，发送方，然后
[k-n,k-1]值的ACK消息由于发送方发送了数据包[k-n,k-1],所以必须是这样的情况:发送方已经接收到用于k-n-1的ACK。一旦接收机发送了AC
它将永远不会发送小于k-n-1的ACK。因此，飞行中的ACK值的范围可以从k-n-1到k-1。

27、

P24. Answer true or false to the following questions and briefly justify your answer:

a. With the SR protocol, it is possible for the sender to receive an ACK for a packet th
outside of its current window. T

b. With GBN, it is possible for the sender to receive an ACK for a packet that falls outs
its current window. T

c. The alternating-bit protocol is the same as the SR protocol with a sender and receiv
window size of 1. T

d. The alternating-bit protocol is the same as the GBN protocol with a sender and rece
window size of 1. T

答案:

a)真。假设发送方具有3的窗口大小并且在T0发送分组1、2、3。在$T1(T1 > T0)$接收机ACK1、2、3。在$t2(t2 > t1)$发送方超时，并且
2、3。在t3,接收机接收重复并重新确认1、2,3、在T4，发送方接收到在T1发送的接收机并使其前进的ACK。窗口至4、5、6。在t5，发送方
送的接收机的ACK1、2、3。这些棚屋在窗户外面。

b)真。实质上与(a)中相同的情形。

c)真。

d)真。请注意，具有1、SR、GBN和交替位协议的窗口大小功能等同。窗口大小1排除了无序分组的可能性(在窗口内)。累积ACK仅仅是普
况，因为它只能指窗口内的单个分组。

28、List six access technologies. Classify each one as home access, enterprise access, or wide-area wireless access.

答案:

1. Dial-u
office; 3.
(802.11):

接入技术	中文名称	分类
1. DSL (Digital Subscriber Line)	数字用户线路	**家庭接入**
2. Cable Internet	电缆互联网	**家庭接入**
3. FTTH (Fiber To The Home)	光纤到户	**家庭/企业接入**
4. Ethernet	以太网	**企业接入**
5. Wi-Fi	无线保真	**家庭/企业接入**
6. 4G/5G (Cellular Network)	第四代/第五代移动通信网络	**广域无线接入**

home or small
5. Wifi

29、**R23. What are the five layers in the Internet protocol stack? What are the principal responsibilities of each of these layers?**

答案：

30、**R25. Which layers in the Internet protocol stack does a router process? Which lay ers does a link-layer switch process? Which layers does a host process?**

答案：

路由器处理网络层、数据链路层和物理层（第 1 层至第 3 层，现代路由器有时会充当防火墙或缓存组件，因此也会处理传输层）。

链路层交换机处理链路层和物理层（第 1 层至第 2 层）。

主机处理所有五层。

31、

R6. In CSMA/CD, after the fifth collision, what is the probability that $K = 4$? The result $K = 4$ corresponds to a delay of how seconds on a 10 Mbps Ethernet?

答案：

在第 5 次碰撞之后，适配器将从{0，1，2，…，31}中选择，它选择 4 的概率是 1/32。需要等待 204.8 微秒。

32、**What does it mean for a wireless network to be operating in “infrastructure mode”? If the network is not in infrastructure mode, what mode of operation is it in, and what is the difference between that mode of operation and infrastructure mode?**

答案：

In infrastructure mode of operation, each wireless host is connected to the larger network via a base station (access point). If not operating in infrastructure mode, a network operates in ad-hoc mode. In ad-hoc mode, wireless hosts have no infrastructure with which to connect. In the absence of such infrastructure, the hosts

themselves must provide for services such as routing, address assignment, DNS-like name translation, and more

在基础设施操作模式中，每个无线主机通过基站（接入点）连接到更大的网络。如果不以基础设施模式运行，则网络以自组织模式运行。在自组织模式下，无线主机没有可连接的基础设施。在没有这样的基础设施的情况下，主机本身必须提供路由、地址分配、类似 DNS 的名称翻译等服务。

为什么802.11无线网络需要确认机制而有线以太网不需要？

33、**Why are acknowledgments used in 802.11 but not in wired Ethernet?**

答案:

无线网络中不能进行冲突检测，因此需要通过确认帧来判断数据是否发送成功，有线网络中可以通过冲突检测机制判断数据是否发送成功，且出错概率小，所以不需要确认帧。

34、Why does collision occur in CSMA, if all nodes perform carrier sensing before transmission?

网络为存在传播延时

如果两个节点同时监听信道同时发送

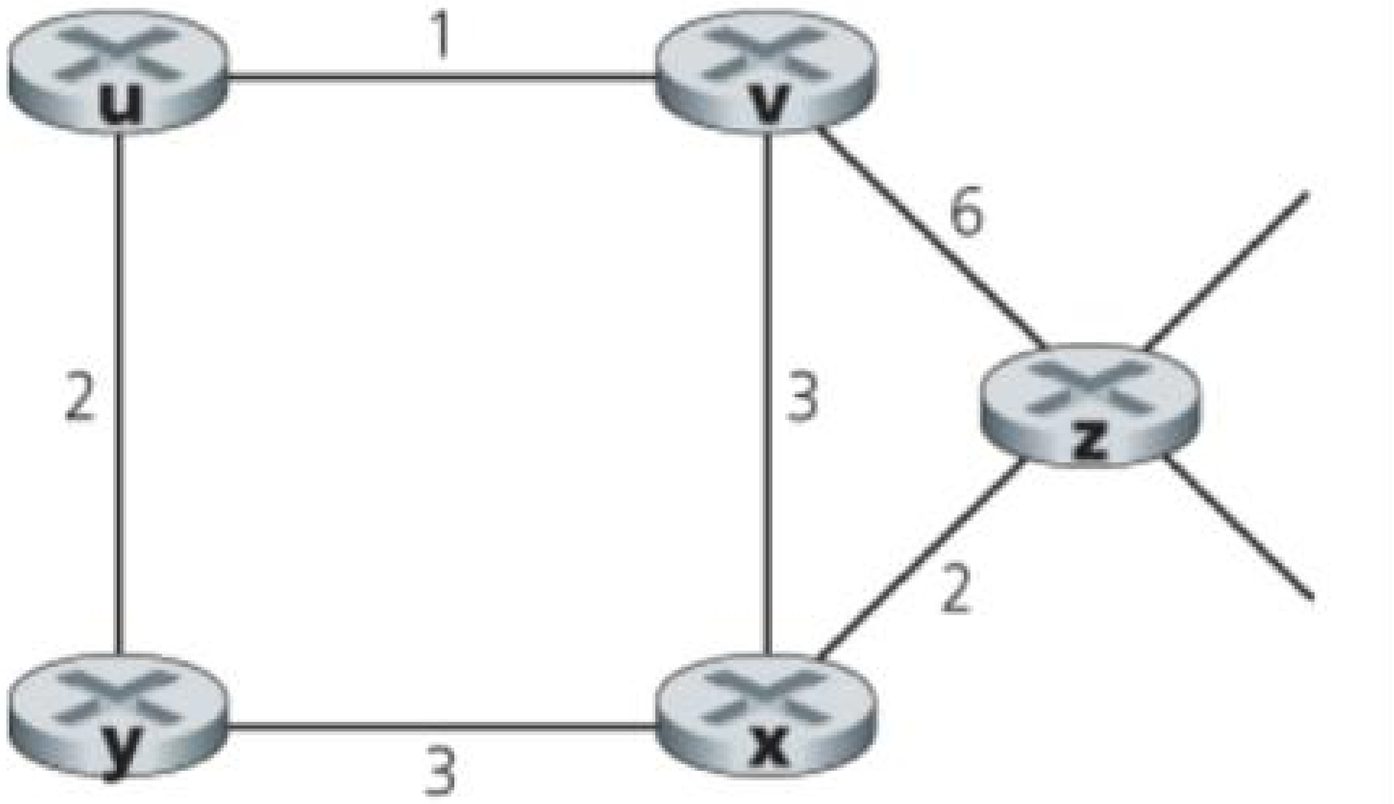
答案:

由于网络存在传播延时，其他节点需要一定时间后才能感知某个节点是否在发送数据，所以即使采用 CSMA，仍然会产生冲突。

35、Consider the network shown below, and assume that each node initially knows the costs to each of its neighbors. Consider the distance-vector algorithm and show the distance table entries at node z.

距离向量算法

	u	v	x	y	z
v	1	0	3	3	5
x	4	3	0	3	2
z	6	5	2	5	0



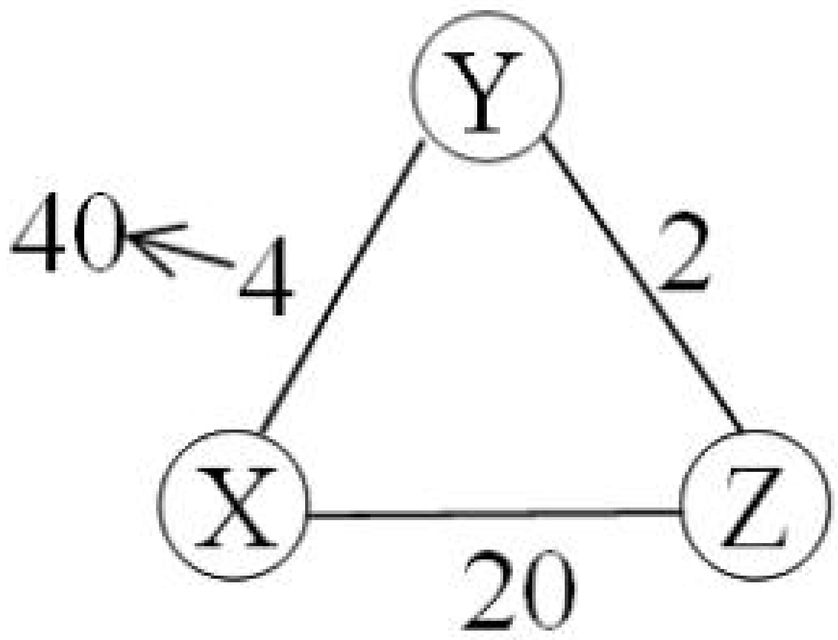
答案:

		值				
		u	v	x	y	z
	v	1	0	3	3	5
从	x	4	3	0	3	2
	z	6	5	2	5	0

36、Consider a three-node topology shown below. If the DV routing algorithm is used in the network, （1） what problem will · happen when

无穷计数问题　　毒性逆转

c(Y,X)· changes· from· 4· to 40?·　(2) In order to solve this problem, after the self-update, what distance vector table does the router Y tell its neighbors? ∞



答案:

1）无穷计数问题　Count to infinite

Y 告诉 Z:

	TO		
	x	y	z
y	∞	0	2

Y 告诉 X:

	TO		
	x	y	z
y	22	0	2

37、Figure 1 shows the network topology, and Figure 2 shows the hex-content of the first 80 bytes of an Ethernet frame of the host for a Web request.(1) What is the IP address of the Web server? What is the MAC address of the default gateway of this host?(2) When the IP packet encapsulated in this frame is forwarded through router R, which fields in the IP packet need to be modified?

目的MAC地址(6Byte)+源MAC地址(6Byte)+类型(2Byte)

(1) 目的IP 40 aa 62 20 即 64.170.98.32

(2) 00 21 27 21 51 ee

修改TTL和首部校验和.



Fig. 1

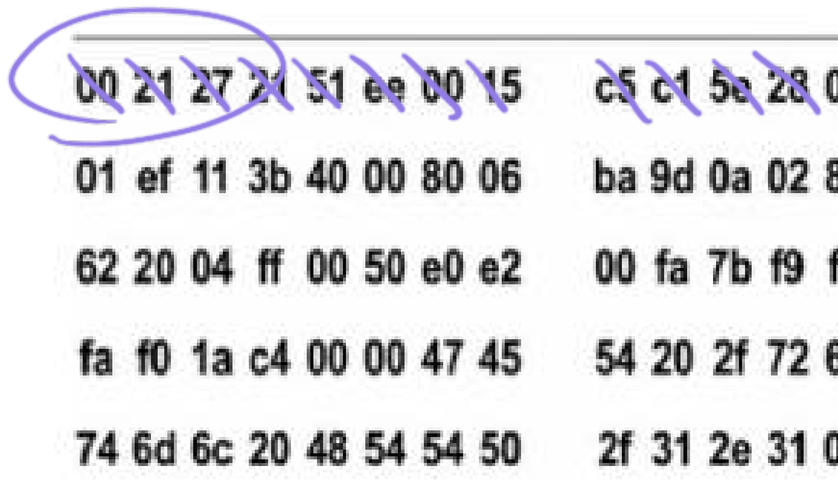


Fig. 2

答案：

(1)64.170.98.32; 00-21-27-21-51-ee

(2) TTL 字段，首部校验和重新计算，总长字段若大于 MTU，则总长字段、标志字段、片偏移字段都需要修改

38、One University owns the block of IP addresses of the form: 103.46.112.0 /20.

Suppose it wants to create four subnets from this block, with each-block having the same number of IP addresses. Please complete the following list.

The block of IP4address	103.46.112.0 /20
The prefixes (of form a.b.c.d/x) for the **lst** subnet	
the prefixes (of form a.b.c.d/x)for the **3rd** subnet	
The directed broadcast address of the **2nd** subnet	
Loop back test address (local host IP address)	

答案：

The block of IP4address	103.46.112.0/20
The prefixes (of form a.b.c.d/x) for the **1st** subnet	103.46.112.0/22
the prefixes (of form a.b.c.d/x)for the **3rd** subnet	103.46.120.0/22
The directed broadcast address of the **2nd** subnet	103.46.119.255
Loop back test address (local host IP address)	127.0.0.1

39、Consider sending a 1600-byte IP datagram into a link that has an MTU of 500 bytes. Assuming a 20-byte IP header and the original datagram is stamped with the identification number 345. Please complete the fragmentation for the large datagram. Give the specific values of the fields related to fragmentation.

答案:

Length=500 ID=345 fragflag=1 offset=0

Length=500 ID=345 fragflag=1 offset=60

Length=500 ID=345 fragflag=1 offset=120

Length=160 ID=345 fragflag=0 offset=180

40、**1. What is the principal responsibility of a router?** _____

2. MTU means _____, it relates to the _____ field in the header of an IP datagram.

3. There are two levels for the routing protocols in the Internet: intra-autonomous system routing protocol and inter-autonomous system routing protocol._____、_____ routing protocols are widely used for intra-AS routing. And all ASs run a same inter-AS routing protocol:_____.

答案:

1. What is the principal responsibility of a router? ____路由和转发_____

2. MTU means__最大传输单元__ ， it relates to the __数据报分段__ field in the header of an IP datagram.

3. There are two levels for the routing protocols in the Internet: intra-autonomous system routing protocol and inter-autonomous system routing protocol.__OSPF___、 __RIP____ routing protocols are widely used for intra-AS routing. And all ASs run a same inter-AS routing protocol:__BGP___.

41、What aspects of the IPv6 protocol has been improved to speed up the IP forwarding speed within the network? The analysis can be combined with the IPv6 and IPv4 datagram formats?

答案:

1）简化头部格式，固定头部长度 2）移除 checksum 字段 3）IPv6 不允许在中间路由器上进行分片与重新组装

42、What two types of ICMP messages are received at the sending host executing the Traceroute program?

答案:

Type 11 code 0 TTL 过期

Type3 code 3 目的端口不可达

在执行 traceout 程序时，发送主机会收到
2种类型的消息 {1 ICMP超时 Type11 code 0
2 ICMP目标不可达 Type3 code 3

目的MAC+目标端口 记录物理映射

43、When does the NAT (Network Address Translation) protocol need? How does it work?

答案:

内网主机与外网主机进行数据传输时，NAT 将内网地址转换成公网 IP 地址，并通过端口号区分不同内网主机，NAT 使用 NAT 路由器上的 NAT 转换表记录公网 IP 地址+端口号与内网 IP+端口号的对应关系。

44、Consider Figure 6.33. Now we replace the router between subnets 1 and 2 with a switch S1, and label the router between subnets 2 and 3 as R1.

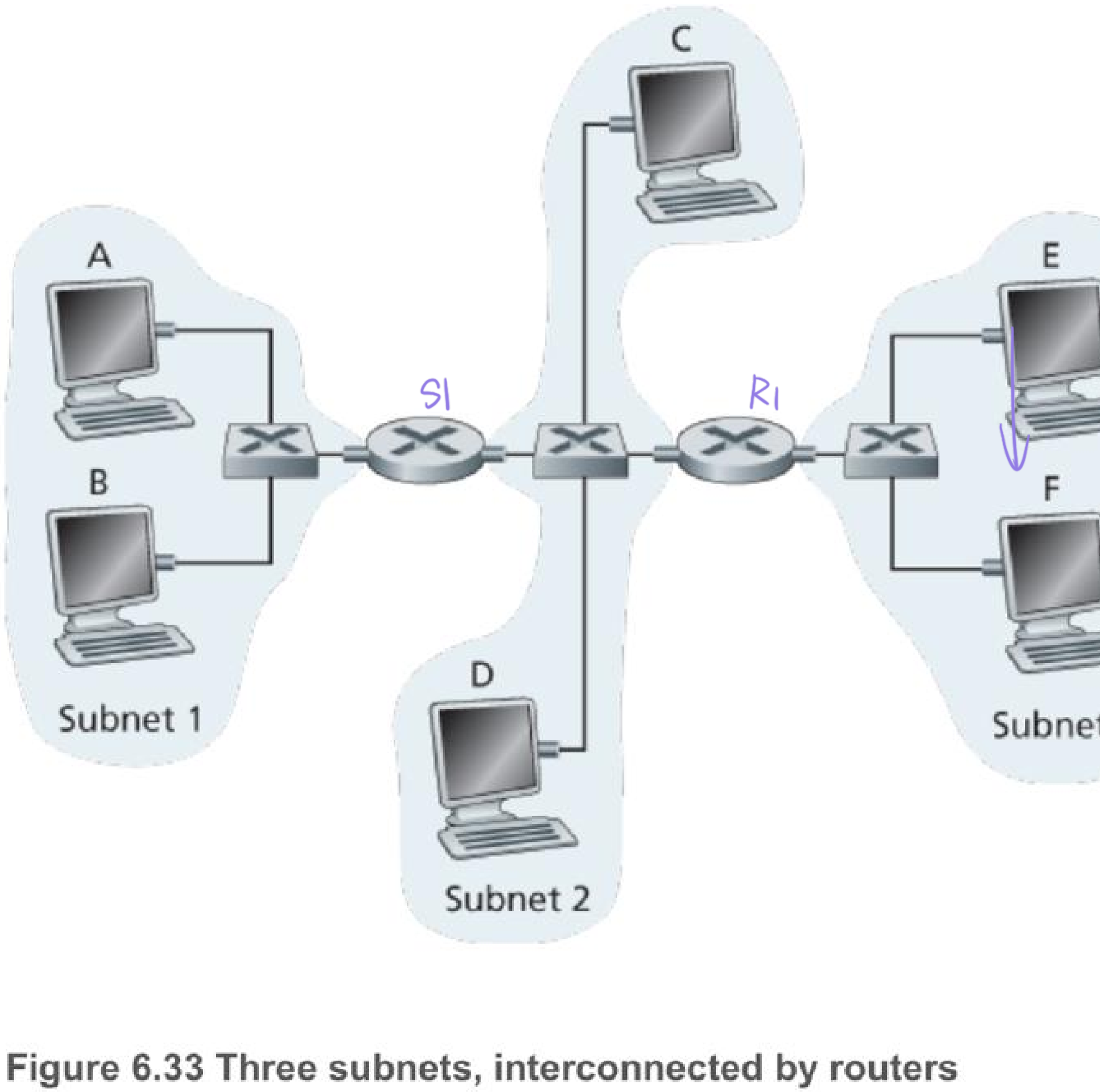


Figure 6.33 Three subnets, interconnected by routers

a. Consider sending an IP datagram from Host E to Host F. Will Host E ask router R1 to help forward the datagram? Why? In the Ethernet frame containing the IP datagram, what are the source and destination IP and MAC addresses?

E和F在一个子网中，可以直接相互通信.

源IP：E的IP

目的IP：F的IP

源MAC：E的MAC

目的MAC：F的MAC

不执行ARP查询，属于不同子网，要将数据包发给默认网关

b. Suppose E would like to send an IP datagram to B, and assume that E's ARP cache does not contain B's MAC address. Will E perform an ARP query to find B's MAC address? Why? In the Ethernet frame (containing the IP datagram destined to B) that is delivered to router R1, what are the source and destination IP and MAC addresses?

源IP：E 目的IP：B 源MAC：E 目的IP：网关的MAC地址

c. Suppose Host A would like to send an IP datagram to Host B, and neither A's ARP cache contains B's MAC address nor does B's ARP cache contain A's MAC address. Further suppose that the switch S1's forwarding table contains entries for Host B and router R1 only. Thus, A will broadcast an ARP request message. What actions will switch S1 perform once it receives the ARP request message? Will router R1 also receive this ARP request message? If so, will R1 forward the message to Subnet 3? Once Host B receives this ARP request message, it will send back to Host A an ARP response message. But will it send an ARP query message to ask for A's MAC address? Why? What will switch S1 do once it receives an ARP response message from Host B?

答案:

a）否。E可以检查主机F的IP地址的子网前缀，然后了解F在同一个局域网上。因此，E不会将数据包发送到默认路由器R1。

从E到F的以太网帧：

源IP=E的IP地址

目的地IP=F的IP地址

源MAC=E的MAC地址

目的地MAC=F的MAC地址

b）不，因为它们不在同一个局域网上。E可以通过检查B的IP地址来发现这一点。

从E到R1的以太网帧：

源IP=E的IP地址

目的地IP=B的IP地址

源MAC=E的MAC地址

目的地MAC=连接到子网的R1接口的MAC地址。

c）交换机S1将通过两个接口广播以太网帧，因为接收到的ARP帧的目的地地址是广播地址。它了解到A驻留在连接到子网1的子网1上，并且S1将更新它的
主机A的条目。

是的，路由器R1也接收这个ARP请求消息，但是R1不会将消息转发到子网3。

B不会发送请求A的MAC地址的ARP查询消息，因为这个地址可以从A的查询消息中获得。

一旦交换机S1接收到B的响应消息，它将在其转发表中为主机B添加一个条目，然后删除接收到的帧，因为目的地主机A与主机B在同一个接口上(即A和B位
段上)。

45、Host A and B are communicating over a TCP connection, and Host B has already received from A all bytes up through byte 126. Suppose Host A then sends two segments to Host B back-to-back. The first and second segments contain 80 and 40 bytes of data, respectively. In the first segment, the sequence number is 127, the source port number is 302, and the destination port number is 80. Host B sends an acknowledgment whenever it receives a segment from Host A.

a. In the second segment sent from Host A to B, what are the sequence number, source port number, and destination port number?

b. If the first segment arrives before the second segment, in the acknowledgment of the first arriving segment, what is the acknowledgment number, the source port number, and the destination port number?

c. If the second segment arrives before the first segment, in the acknowledgment of the first arriving segment, what is the acknowledgment number?

答案:

a) 在从主机 A 到 B 的第二段中，序列号为 207，源端口号为 302，目的端口号为 80。

b) 如果所述第一段在所述第二到达段之前到达，则在所述第一到达段的确认中，所述确认号为 207，所述源端口号为 80，所述目的端口号为 302。

c) 如果第二段到达第一段之前，在第一个到达段的确认中，确认号为 127，表示它仍在等待字节 127 和以后。

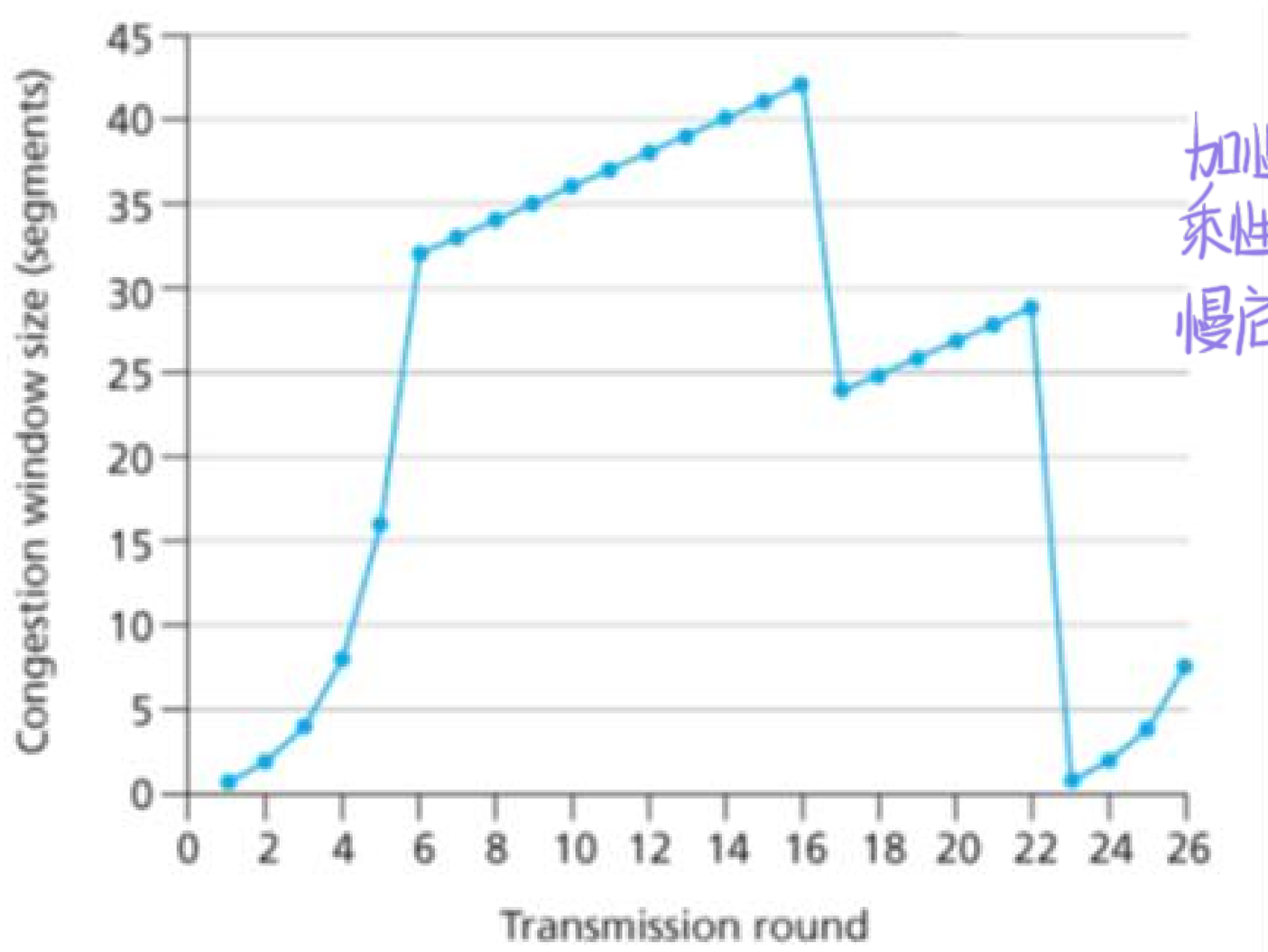
46、Host A and B are directly connected with a 100 Mbps link. There is one TCP connection between the two hosts, and Host A is sending to Host B an enormous file over this connection. Host A can send its application data into its TCP socket at a rate as high as 120 Mbps but Host B can read out of its TCP receive buffer at a maximum rate of 50 Mbps. Describe the effect of TCP flow control.

答案:

由于链路容量只有 100 Mbps，所以主机 A 的发送速率最多可达 100 Mbps。不过，主机 A 向接收缓冲区发送数据的速度比主机 B 从缓冲区中删除数据的速

度要快。接收缓冲区的填充速率约为 40 Mbps。当缓冲区已满时，主机 B 通过设置 RcvWindow=0 向主机 A 发送停止发送数据的信号。然后主机 A 停止发送，直到接收到 RcvWindow>0 的 TCP 段为止。主机 A 将作为 RcvWindow 值的函数反复停止并开始发送从主机 B 接收。平均而言，主机 A 向主机发送数据的长期速率 b 作为此连接的一部分，不超过 60Mbps.

47、Consider Figure 3.58 . Assuming TCP Reno is the protocol experiencing the behavior shown above, answer the following questions. In all cases, you should provide a short discussion justifying your answer.



加性增
乘性减
慢启动

TCP Reno 拥塞控制核心阶段回顾

1. **慢启动（Slow Start）**：cwnd 从 1 开始，每轮次指数级增长（$\text{cwnd} \times 2$），直到 cwnd 达到**慢启动阈值（ssthresh）**。
2. **拥塞避免（Congestion Avoidance）**：cwnd 达到 ssthresh 后，每轮次线性增长（$\text{cwnd} + 1$）。
3. **快速重传与快速恢复（Fast Retransmit & Fast Recovery）**：收到 3 个重复 ACK 时，ssthresh 减半，cwnd 设为新的 ssthresh，进入快速恢复阶段（cwnd 线性增长）；若发生超时，则 ssthresh 减半，cwnd 重置为 1，重新慢启动。

逐阶段分析图中 cwnd 变化

阶段 1：慢启动与拥塞避免（轮次 0~16）

- 初始 cwnd=1，轮次 0~5 期间，cwnd 指数级增长（符合慢启动）。
- 当 cwnd 达到**ssthresh=32**（轮次 6 左右）后，进入拥塞避免阶段，cwnd 线性增长，直到轮次 16 达到峰值（约 42）。

阶段 2：第一次拥塞事件（轮次 16）

- 轮次 16 时 cwnd 骤降，说明发生**超时重传**（而非 3 个重复 ACK，因为若为快速重传，cwnd 不会直接重置为 1）。
- 处理：ssthresh 减半为$42/2 = 21$，cwnd 重置为 1，重新进入慢启动。

阶段 3：慢启动与拥塞避免（轮次 17~22）

- cwnd 从 1 开始指数增长，直到达到新的 ssthresh=21（轮次 19 左右），随后线性增长，直到轮次 22。

阶段 4：第二次拥塞事件（轮次 22）

- 轮次 22 时 cwnd 再次骤降，同样是**超时重传**。
- 处理：ssthresh 减半为$30/2 = 15$（假设轮次 22 前 cwnd 约为 30），cwnd 重置为 1，重新慢启动（轮次 23~26）。

operating.

triple duplicate ACK

a triple duplicate ACK

nd?

a. Identify the intervals of time when TCP slow start is operating.

0-6

b. Identify the intervals of time when TCP congestion avoidance is operating.

6-16

c. After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?

3 次重复的 ACK，进入快速重传

d. After the 22nd transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?

Timeout

e. What is the initial value of ssthresh at the first transmission round?

Ssthresh=32

48、In the OSI model, the first bottom-up level to provide end-to-end services is (B)

B.

A、 data link layer

OSI模型中自底向上第一个提供端到端服务的层次是传输层.

B、 transport layer

C、 session layer

D、 application layer

答案： B

49、Among the following options, which is not described in the network architecture is (C)

网络层

网络架构不涉及具体技术实现细节

A、 layers in the network

B、 protocols used by each layer

C、 internal implementation details of the protocols

D、 functions that must be completed by each layer

答案： C

50、In the TCP / IP reference model, what directly provides services for the application layer is ( )

B 传输层为应用层直接提供服务

A、 Presentation layer

B、 Transmission layer

C、 Network layer

D、

Network interface layer

答案： B

51、Assuming that the application layer in the ISO/OSI reference model wants to send 400B data (without splitting), except the physical layer and application layer, all other layers introduce 20B additional overhead when packaging PDU, the transmission efficiency of the application layer is about ()

A

A、 80%

$\frac{400}{400+100}=80\%$

B、 83%

C、 87%

D、 91%

答案： A

52、In the physical layer interface characteristics, the time sequence used to describe the completion of each function is （)

A、

Mechanical （机械）Characteristics

B、

Electrical （电气）Characteristics

C、

Functional （功能）Characteristics

D、 用于描述完成各项功能的时间序列

Procedural（规程） Characteristics

答案： D

53、Among the following options, that do not belong to the definition scope of physical layer interface specification is （）

A、 interface shape 机械特性

B、 pin function 功能特性

C、 physical address ⇒数据链路层范畴

D、 signal level 电气特性

答案： C

54、Under the condition of no noise, if the frequency bandwidth of a communication link is 3kHz and QAM modulation technology with 4 phases and 4 amplitudes in each phase is adopted, the maximum data transmission rate of the communication link is （） B

A、 在没有噪声的情况下，应用奈奎斯特定律：

12kbps $C = 2B\log_2(M)$

B、 $M = 4\times4 = 16$

24kbps $B = 3000$

$C = 6000 \times \log_2 16 = 24000 = 24kbps$

C、

48kbps

D、

96kbps

答案： B

解析： 波特率 6K，每个波特表示 4bit。C=2Wlog2N=2 * 3K * log2 （4*4）=24Kbps

55、 If the data transmission rate of a communication link is 2400bps and 4-phase modulation is adopted, the baud rate of the link is （）

A、

600 baud

比特率：波特率×每个码元的比特数

B、

则 $B=\frac{2400}{2}=1200$ baund

1200 baud

（4相意味着每个码元 $(2^2=4)$ 可以表示2个bit）

C、

4800 baud

D、

9600 baud

答案： B

解析：

2. （11 年考研 34 题）若某通信链路的数据传输速率为 2400bps，采用 4 相位调制，则该链路的波特率是 B

A. 600 波特　　B. 1200 波特　　C. 4800 波特　　D. 9600 波特

【解答】选 B。有 4 种相位，则一个码元需要由 $\log_2 4=2$ 个 bit 表示，则波特率=比特率/2=1200 波特。

【考查知识点】考察波特与比特率的关系；比特率和波特率之间的换算关系如下：比特率 = 波特率 * $\log_2 n$ 。比特率是数字信号的传输速率——单位时间内所传输的二进制代码的有效位数。单位——比特/秒（bps）或千比特/秒（kbps）。波特率是调制速率（又称波形速率）——线路中每秒传送的波形的个数。单位——波特（band）。

56、If the frequency bandwidth connecting R2 and R3 link is 8kHz and the signal-to-noise ratio is 30dB, and the actual data transmission rate of the link is about 50% of the

信噪比 $10\log\frac{S}{N}=30$ 则 $\frac{S}{N}=1000$

$C=B\log_2(\frac{S}{N}+1)=8\times\log_2 1001=80$ kbps

maximum data transmission rate, the actual data transmission rate of the link is about ()

A、 8kbps

B、 20kbps

C、 40kbps

D、 80kbps

答案： C
解析：

根据香农公式：

理论最大：8klog2（1+1000）=8k10=80kbps

实际最大：0.5*理论最大=40kbps

57、Among the following factors, which will not affect the channel data transmission rate is ( )

A、 signal to noise ratio

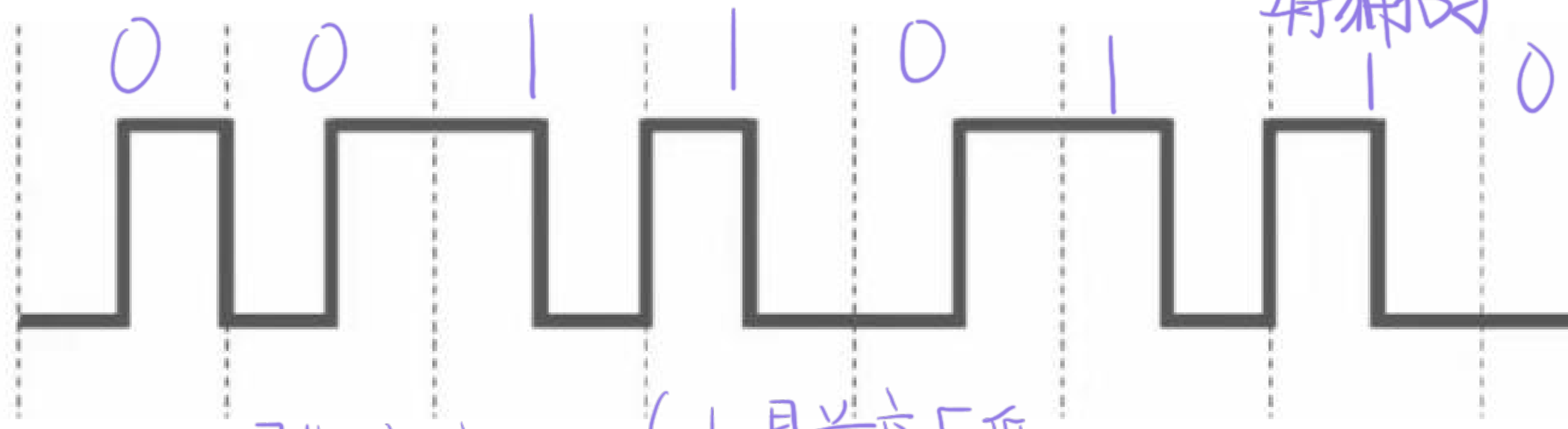
B、 frequency bandwidth

C、 symbol speed

D、 signal propagation speed

答案： D

58、If the following figure shows the signal waveform received by the 10Base-T network interface card, the bit string received by the network interface card is ()



A、 00110110

B、 10101101

C、 01010010

D、 11000101

答案： A
解析：从波形看出是曼彻斯特编码，所以是 00110110 或 11001001

59、The data link layer uses the GBN protocol, and the sender has sent frames numbered 0 ~ 7. When the timer expires, if the sender only receives the acknowledgement of frames 0, 2 and 3, the number of frames to be retransmitted by the sender is ( )

A、 2

B、 3

C、 4

D、 5

答案： C
解析：
已经收到 3 号帧的确认，表示 0—3 号帧都已经正确接收，超时发生就重发 4—7 帧。GBN 不只是超时重传的特点，它还有累计确认的功能，即：发一个确认 N，表示前 N 帧都已正确收到。超时重传一旦发生，就重发最后确认帧之后的所有帧。不要和选择重传（SR）弄混，选择重传可以乱序接收，只发送单个帧的确认。如果是选择重传，还要再多传 1 号帧。

60、In a LAN using CSMA / CD protocol, the transmission medium is a complete cable, the transmission rate is 1Gbps, and the signal propagation rate in the cable is 200000 km / s. if the minimum data frame length is reduced by 800 bits, the distance between the farthest two stations needs to be at least ( )

A、

increased by 160m

B、

increased by 80m

C、

reduced by 160m

D、

reduced by 80m

答案： D
解析：

解析：

1.传输时延>2*最远两个站点间的传播时延

2.传输时延=数据包的大小/数据包的发送率

3.传播时延=站点间距离/信号传播速率

1 式为最远站点发出的数据产生冲突后，能被远端站点感知的最短时间，保证了在远端站点发送完数据之前能够检测到冲突

由 2 式，数据帧减小 800bit，即传输时延减小 800bit/1Gbps=8*10^(-7) s

由 1 式，因此传播时延至少减小 1/2 * 8*10^(-7) =4*10^(-7) s

由 3 式，因此站点间距离最少减少 4*10^(-7) * 200000 = 0.08km = 80m

最短帧长 = 2 * 传播时延 * 数据传输率

$$800\,bit = 2 * \frac{d}{200000\,km/s} * 1\,Gb/s$$

$$= 2 * \frac{d}{2 \times 10^8 m/s} * 10^9\,bit/$$

$$10d = 800\,m$$

$$d = 80\,m$$

故需减少 80m 的站点距离。

61、Host A sends a TCP segment (SYN=1,seq=11220) to host B and expects to establish a TCP connection with host B. if host B accepts the connection request, the correct TCP segment sent by host B to host A may be (C)

SYN=1 seq=11221

ACK=11221

A、 SYN=0,ACK=0,seq=11221,ack=11221

B、

SYN=1,ACK=1,seq=11220,ack=11220

C、

SYN=1,ACK=1,seq=11221,ack=11221

D、 SYN=0,ACK=0,seq=11220,ack=11220

答案： C

62、A TCP connection is established between host A and host B, and the MSS length of TCP is 1000 bytes. If the current congestion window of host A is 4000 bytes, after host A sends two MSS segments to host B continuously, host A successfully receives the confirmation segment of the first segment sent by host B, and the size of the receiving window in the confirmation segment is 2000 bytes, then the maximum number of bytes that host A can also send to host B is ( )

A、 1000

B、 2000

C、 3000

D、 4000

答案： A

63、A TCP connection has been established between host A and host B. host A sends three consecutive TCP segments to host B, including 300 bytes, 400 bytes and 500 bytes payload respectively. The sequence number of the third segment is 900. If host B receives only the first and third segments correctly, the confirmation sequence number sent by host B to host A is( )

A、 300

B、 500

C、 1200

D、 1400

答案： B

64、A TCP connection always sends TCP segments with 1KB MSS. The sender has enough data to send. A timeout event occurs when the congestion window is 16KB. If the transmission of TCP segments in the next four RTTS is successful, when all TCP segments sent in the fourth RTT receive positive responses, the size of the congestion window is( )

A、 7KB

B、 8KB

C、 9KB

解析：TCP 超时后，拥塞窗口重置为 1KB，慢启动阈值设为 8KB。

- 第 1 个 RTT：拥塞窗口从 1KB 增长到 2KB；
- 第 2 个 RTT：增长到 4KB；
- 第 3 个 RTT：增长到 8KB（达到慢启动阈值，进入拥塞避免阶段，每次 RTT 拥塞窗口增加 1KB）；
- 第 4 个 RTT：拥塞窗口从 8KB 增长到 9KB。

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慢启动阈值为什么是8

在 TCP 拥塞控制中，当发生超时事件时，慢启动阈值（ssthresh）会被设置为**当前拥塞窗口的一半**。

本题中，超时发生时拥塞窗口是 16KB，因此慢启动阈值 $\text{ssthresh} = 16 \div 2 = 8\text{KB}$。

D、 16KB

答案： C

65、A IPv6 address with zero compression is 1A00::C057:50:8:15/60, the full expression of this address with colon hexadecimal notation is_____.

答案：

1A00:0:0:0:C057:50:8:15/60

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